Arboring Widths - Angiosperms

Tree ID Time Series for Angiosperms

```
#reshape data from wider to longer format

library(tidyverse)
```

```
Warning: package 'tidyverse' was built under R version 4.5.2

Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'tibble' was built under R version 4.5.2

Warning: package 'tidyr' was built under R version 4.5.2

Warning: package 'readr' was built under R version 4.5.2

Warning: package 'purrr' was built under R version 4.5.2

Warning: package 'dplyr' was built under R version 4.5.2

Warning: package 'forcats' was built under R version 4.5.2

Warning: package 'lubridate' was built under R version 4.5.2

── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.1.4     ✓ readr     2.1.6
✓ forcats   1.0.1     ✓ stringr   1.5.2
✓ ggplot2   4.0.1     ✓ tibble    3.3.0
✓ lubridate 1.9.5     ✓ tidyr     1.3.2
✓ purrr     1.2.1
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✗ dplyr::filter() masks stats::filter()
✗ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
errors
```

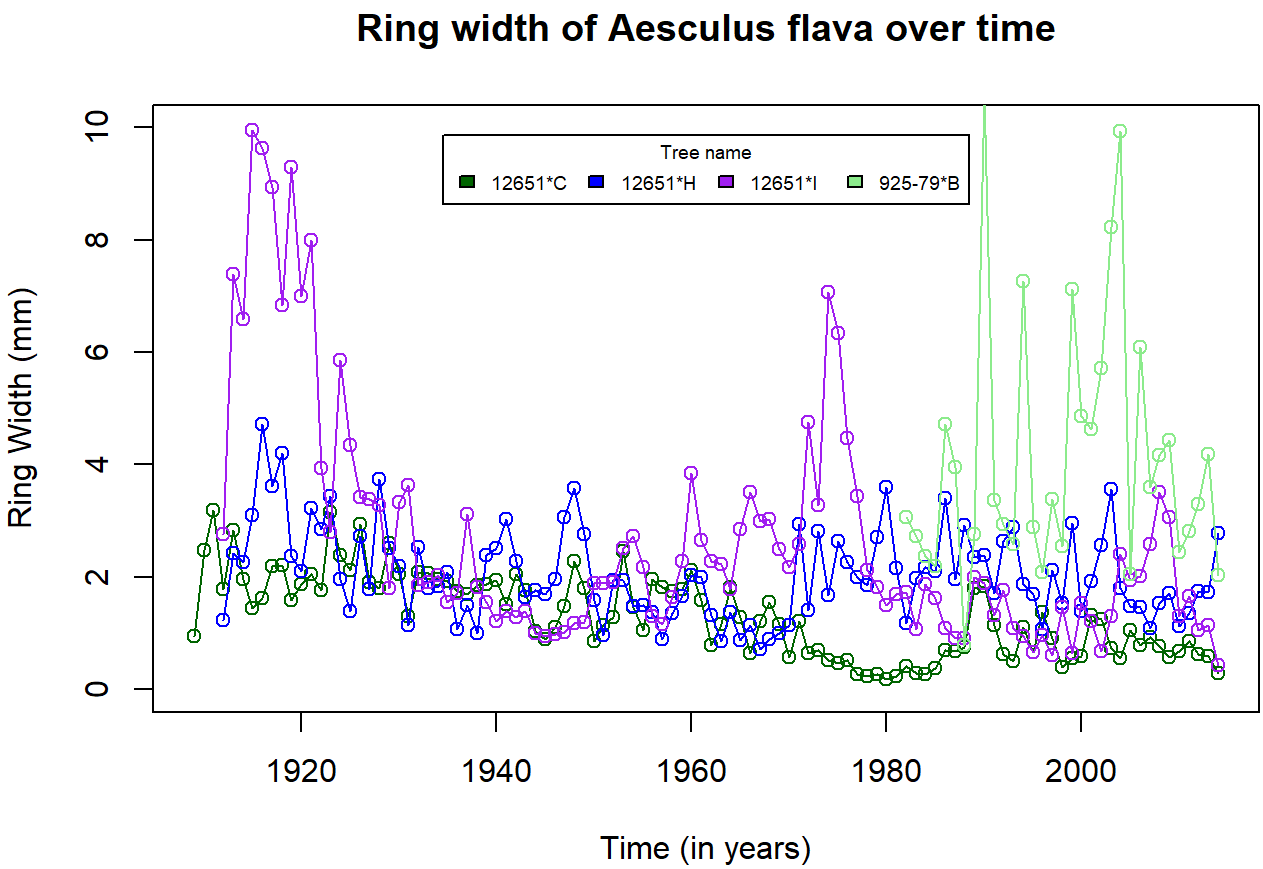
```
# Load data
angio <- read.csv("Arboringwidthsangiosperms.csv")

# Remove X in front of year columns by subbing "X" with nothing ""
colnames(angio) <- sub("X", "", colnames(angio))

# Pivot data
angio_data_long <- pivot_longer(data = angio, cols = "2014":"1802", names_to = "Year", values_to =
```

```r
# View data
print(angio_data_long)
```

```
# A tibble: 14,056 × 12
   TreeName Site  YearLastRing Species RingCount DataType OffsetToNext ImageName
   <chr>    <chr>        <int> <chr>       <int> <chr>           <int> <chr>
 1 12651*C  Arn …         2014 Aescul…       107 RINGWID…            1 Aesculus…
 2 12651*C  Arn …         2014 Aescul…       107 RINGWID…            1 Aesculus…
 3 12651*C  Arn …         2014 Aescul…       107 RINGWID…            1 Aesculus…
 4 12651*C  Arn …         2014 Aescul…       107 RINGWID…            1 Aesculus…
 5 12651*C  Arn …         2014 Aescul…       107 RINGWID…            1 Aesculus…
 6 12651*C  Arn …         2014 Aescul…       107 RINGWID…            1 Aesculus…
 7 12651*C  Arn …         2014 Aescul…       107 RINGWID…            1 Aesculus…
 8 12651*C  Arn …         2014 Aescul…       107 RINGWID…            1 Aesculus…
 9 12651*C  Arn …         2014 Aescul…       107 RINGWID…            1 Aesculus…
10 12651*C  Arn …         2014 Aescul…       107 RINGWID…            1 Aesculus…
# ℹ 14,046 more rows
# ℹ 4 more variables: Analysis.Date.Time <chr>, PathLength <dbl>, Year <chr>,
#   Ringwidth <dbl>
```

Aesculus flava

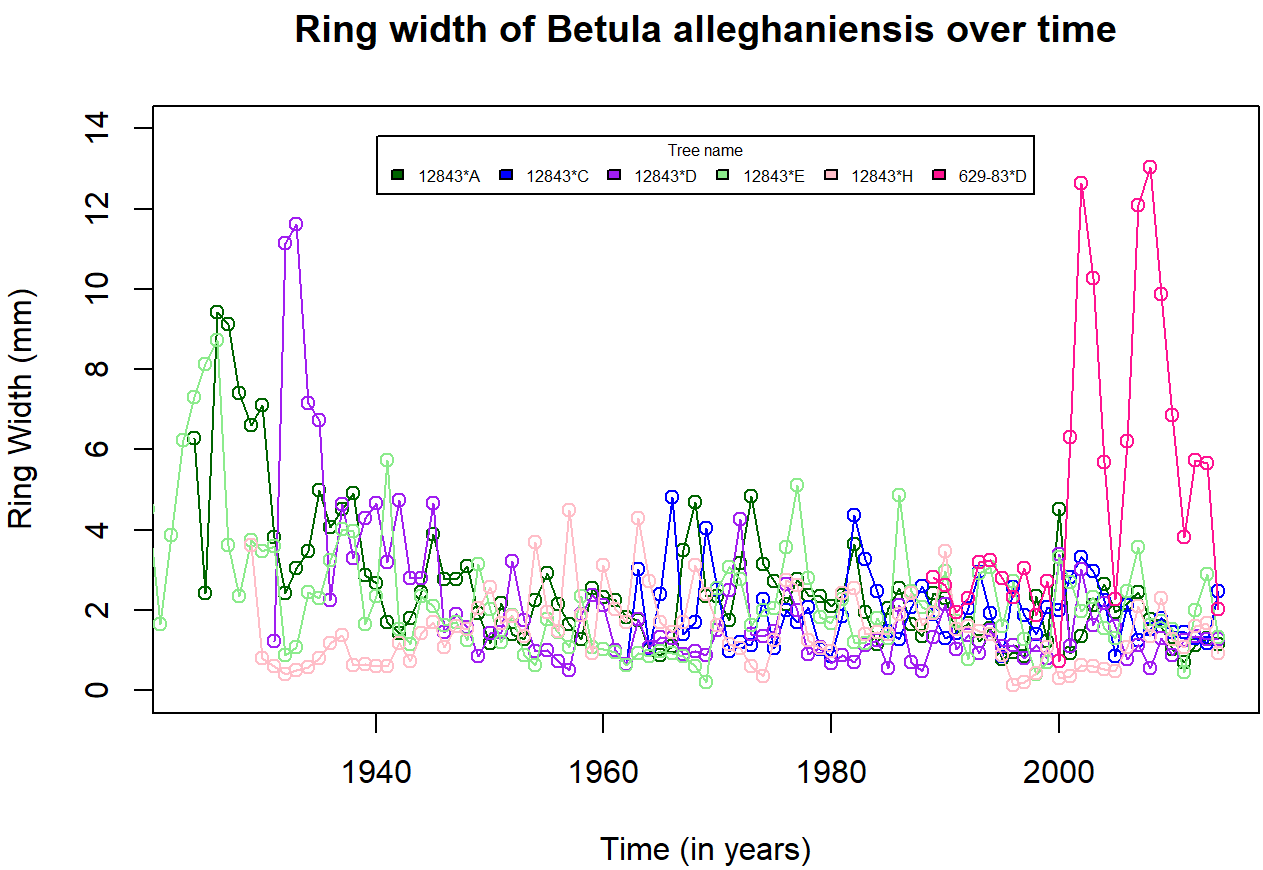
```r
flava1 <- subset(angio_data_long, TreeName == "12651*C")

flava2 <- subset(angio_data_long, TreeName == "12651*H")

flava3 <- subset(angio_data_long, TreeName == "12651*I")

flava4 <- subset(angio_data_long, TreeName == "925-79*B")


plot(flava1$Year, y = flava1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(flava2$Year, y = flava2$Ringwidth, type = "o", col = "blue")
lines(flava3$Year, y = flava3$Ringwidth, type = "o", col = "purple")
lines(flava4$Year, y = flava4$Ringwidth, type = "o", col = "lightgreen")

legend("top", inset = .05, title = "Tree name", c("12651*C", "12651*H", "12651*I", "925-79*B"), f
```



Betula alleghaniensis

```
alle1 <- subset(angio_data_long, TreeName == "12843*A")

alle2 <- subset(angio_data_long, TreeName == "12843*C")

alle3 <- subset(angio_data_long, TreeName == "12843*D")

alle4 <- subset(angio_data_long, TreeName == "12843*E")

alle5 <- subset(angio_data_long, TreeName == "12843*H")

alle6 <- subset(angio_data_long, TreeName == "629-83*D")


plot(alle1$Year, y = alle1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(alle2$Year, y = alle2$Ringwidth, type = "o", col = "blue")
lines(alle3$Year, y = alle3$Ringwidth, type = "o", col = "purple")
lines(alle4$Year, y = alle4$Ringwidth, type = "o", col = "lightgreen")
lines(alle5$Year, y = alle5$Ringwidth, type = "o", col = "pink")
lines(alle6$Year, y = alle6$Ringwidth, type = "o", col = "deeppink")
```

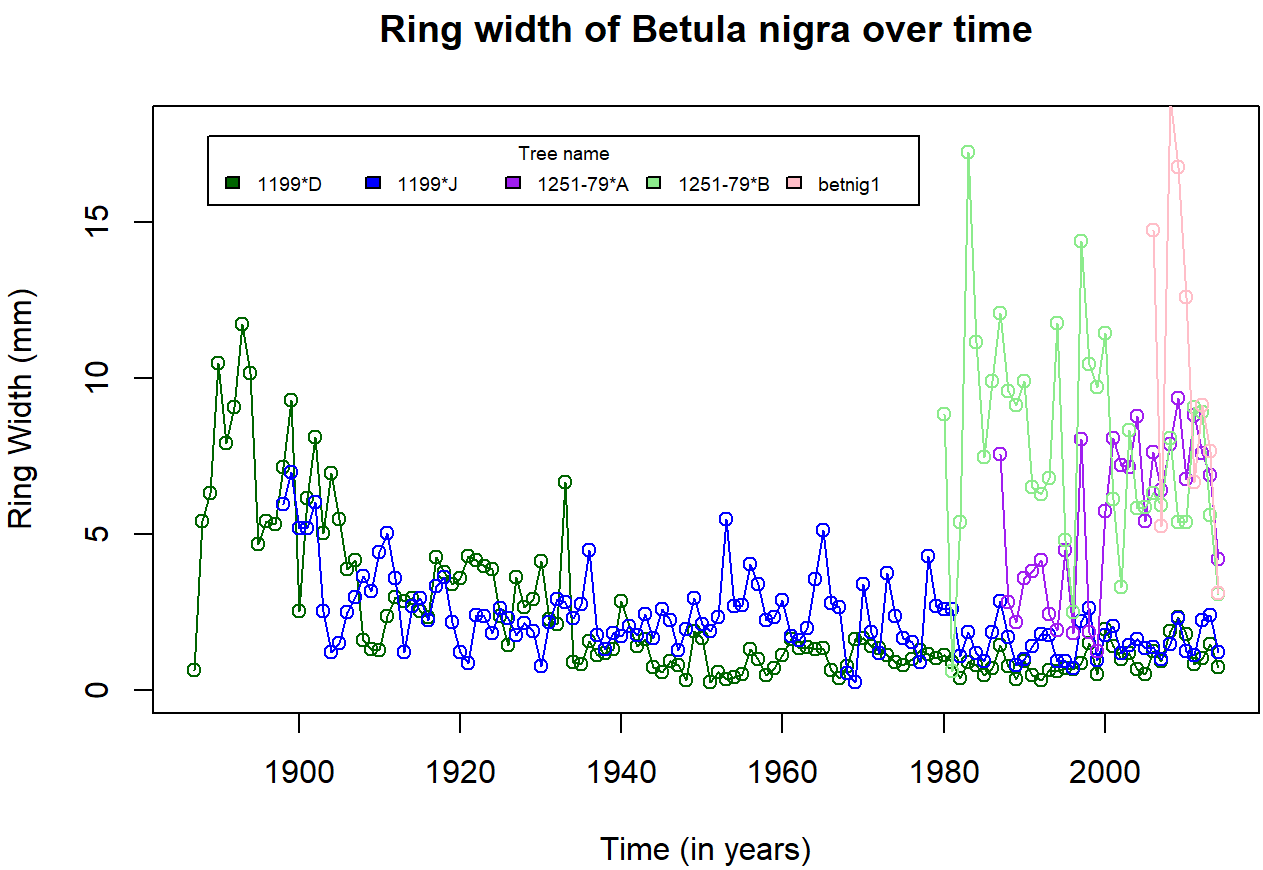
```
legend("top", inset = .05, title = "Tree name", c("12843*A", "12843*C", "12843*D", "12843*E", "12
```



Betula nigra

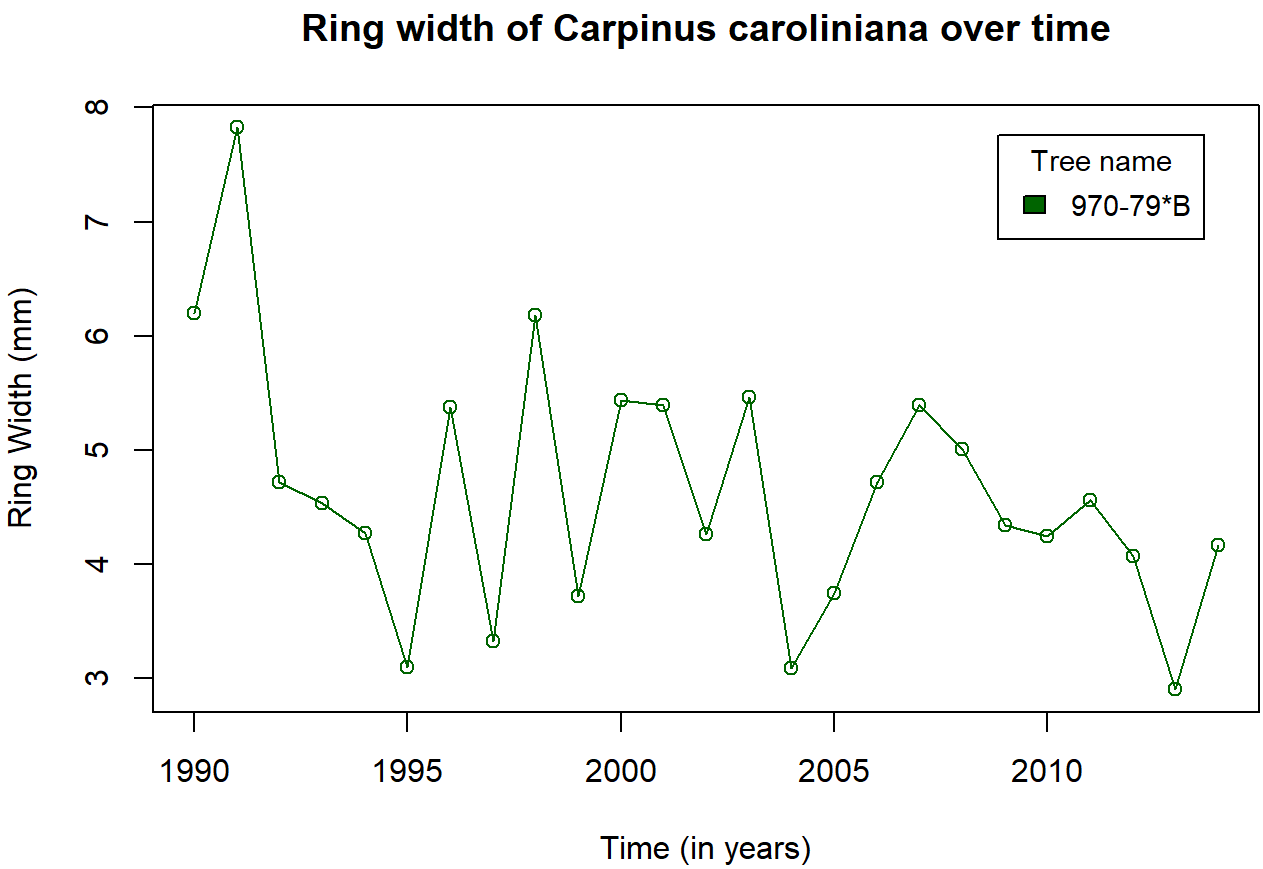
```
betn1 <- subset(angio_data_long, TreeName == "1199*D")

betn2 <- subset(angio_data_long, TreeName == "1199*J")

betn3 <- subset(angio_data_long, TreeName == "1251-79*A")

betn4 <- subset(angio_data_long, TreeName == "1251-79*B")

betn5 <- subset(angio_data_long, TreeName == "betnig1")


plot(betn1$Year, y = betn1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(betn2$Year, y = betn2$Ringwidth, type = "o", col = "blue")
lines(betn3$Year, y = betn3$Ringwidth, type = "o", col = "purple")
lines(betn4$Year, y = betn4$Ringwidth, type = "o", col = "lightgreen")
lines(betn5$Year, y = betn5$Ringwidth, type = "o", col = "pink")
```

```
legend("topleft", inset = .05, title = "Tree name", c("1199*D", "1199*J", "1251-79*A", "1251-79*B
```



Carpinus caroliniana

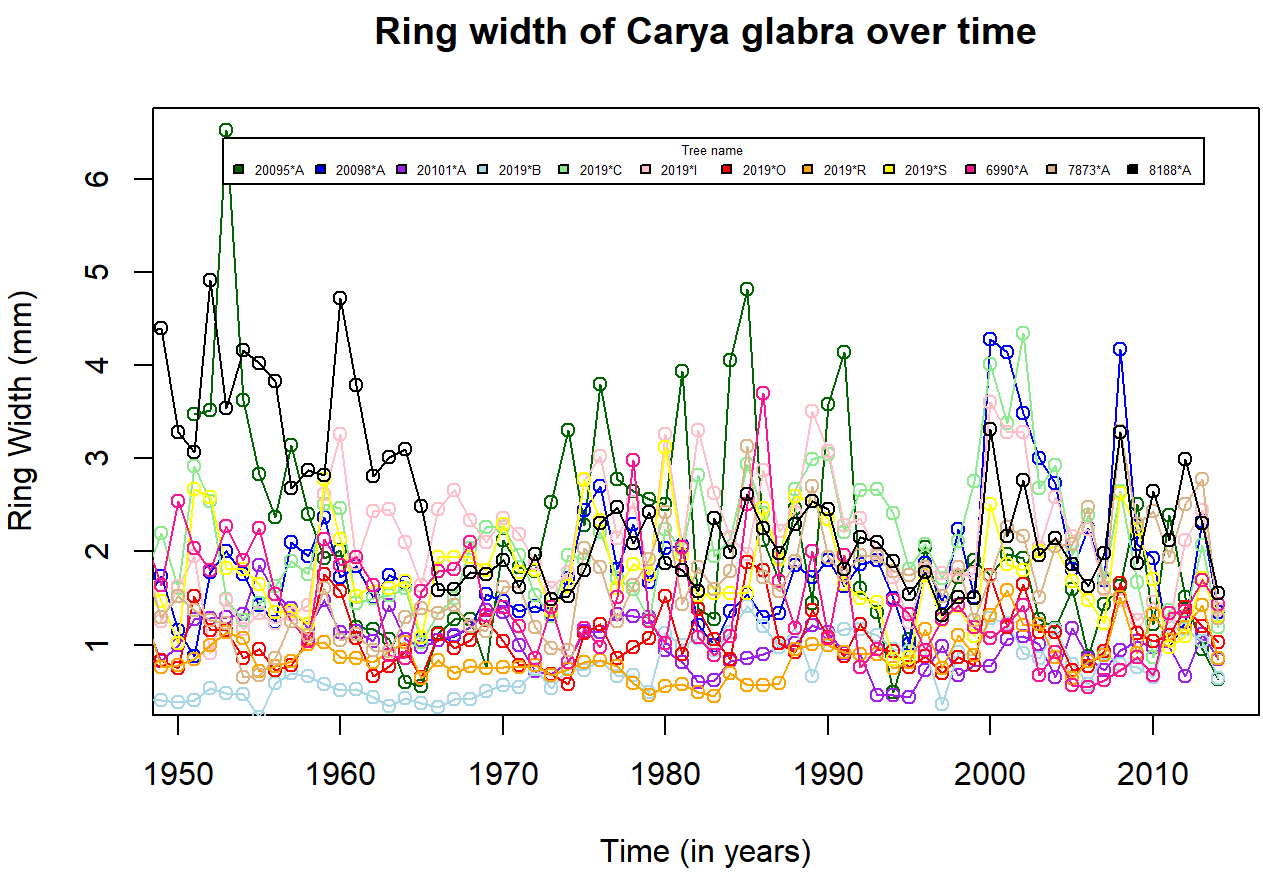
```
carol1 <- subset(angio_data_long, TreeName == "970-79*B")


plot(carol1$Year, y = carol1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal

legend("topright", inset = .05, title = "Tree name", c("970-79*B"), fill = c("darkgreen"), horiz
```



Carya glabra

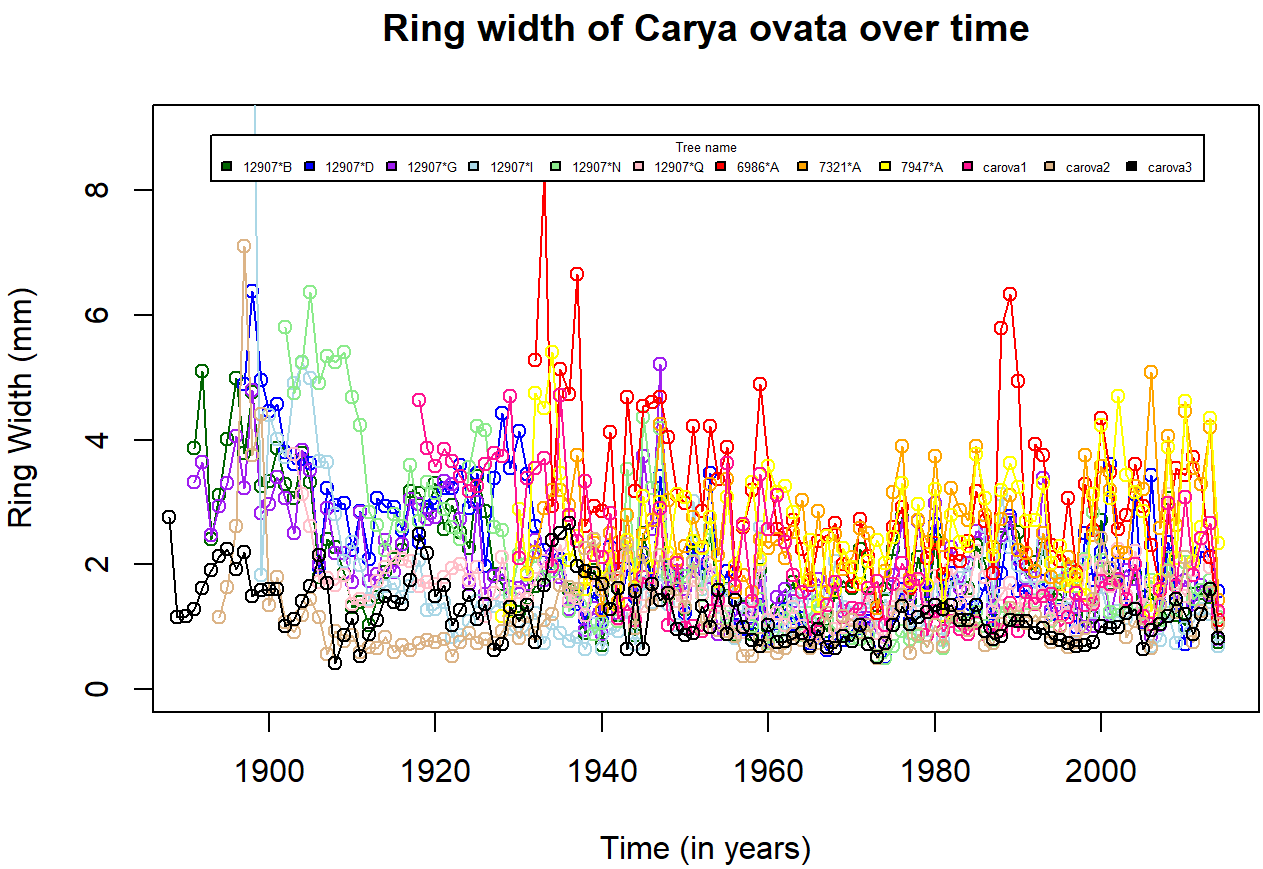
```
glab1 <- subset(angio_data_long, TreeName == "20095*A")

glab2 <- subset(angio_data_long, TreeName == "20098*A")

glab3 <- subset(angio_data_long, TreeName == "20101*A")

glab4 <- subset(angio_data_long, TreeName == "2019*B")

glab5 <- subset(angio_data_long, TreeName == "2019*C")

glab6 <- subset(angio_data_long, TreeName == "2019*I")

glab7 <- subset(angio_data_long, TreeName == "2019*O")

glab8 <- subset(angio_data_long, TreeName == "2019*R")

glab9 <- subset(angio_data_long, TreeName == "2019*S")

glab10 <- subset(angio_data_long, TreeName == "6990*A")

glab11 <- subset(angio_data_long, TreeName == "7873*A")
```

```
glab12 <- subset(angio_data_long, TreeName == "8188*A")


plot(glab1$Year, y = glab1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(glab2$Year, y = glab2$Ringwidth, type = "o", col = "blue")
lines(glab3$Year, y = glab3$Ringwidth, type = "o", col = "purple")
lines(glab4$Year, y = glab4$Ringwidth, type = "o", col = "lightblue")
lines(glab5$Year, y = glab5$Ringwidth, type = "o", col = "lightgreen")
lines(glab6$Year, y = glab6$Ringwidth, type = "o", col = "pink")
lines(glab7$Year, y = glab7$Ringwidth, type = "o", col = "red")
lines(glab8$Year, y = glab8$Ringwidth, type = "o", col = "orange")
lines(glab9$Year, y = glab9$Ringwidth, type = "o", col = "yellow")
lines(glab10$Year, y = glab10$Ringwidth, type = "o", col = "deeppink")
lines(glab11$Year, y = glab11$Ringwidth, type = "o", col = "burlywood")
lines(glab12$Year, y = glab12$Ringwidth, type = "o", col = "black")

legend("topright", inset = .05, title = "Tree name", c("20095*A", "20098*A", "20101*A", "2019*B",
```



Carya ovata

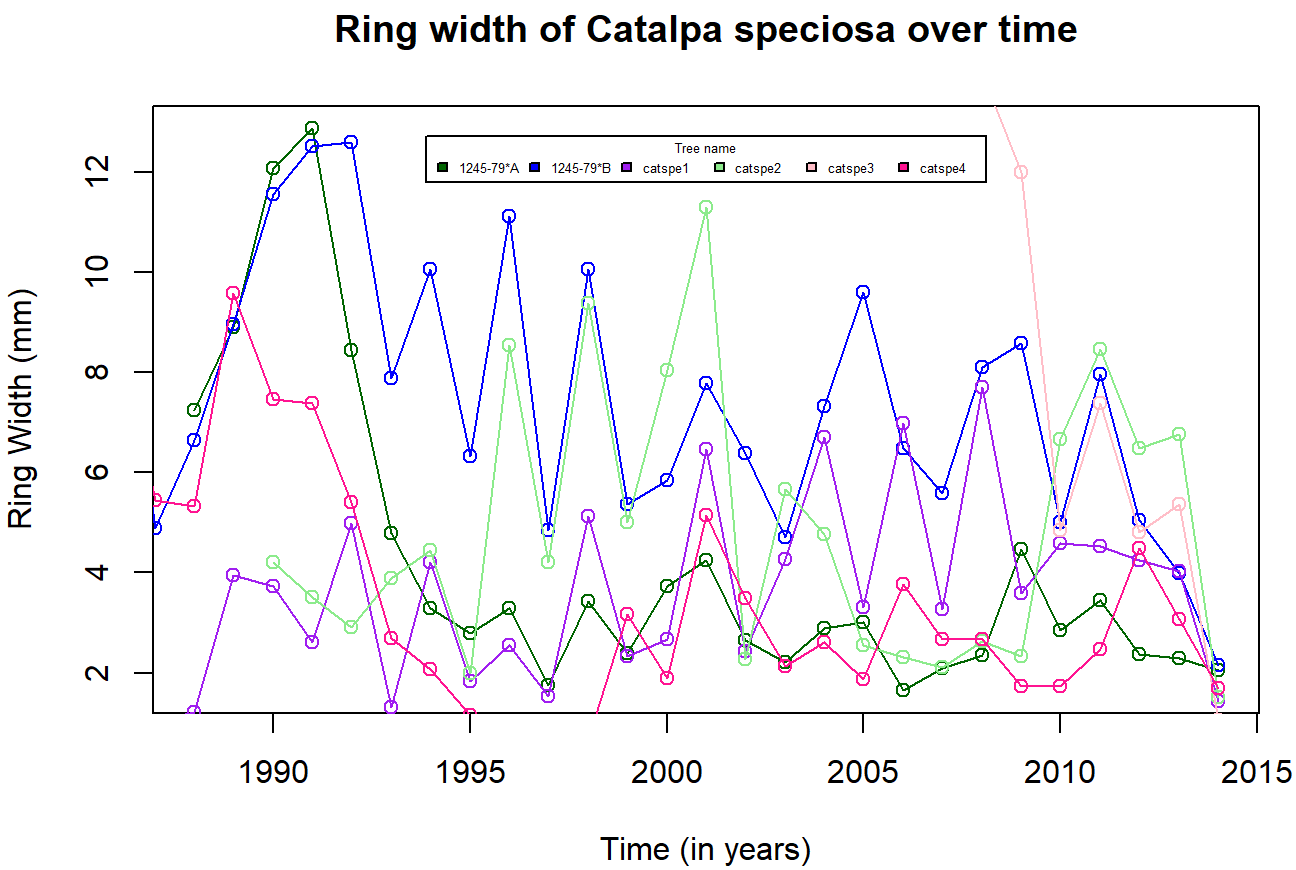
```
ov1 <- subset(angio_data_long, TreeName == "12907*B")

ov2 <- subset(angio_data_long, TreeName == "12907*D")

ov3 <- subset(angio_data_long, TreeName == "12907*G")

ov4 <- subset(angio_data_long, TreeName == "12907*I")

ov5 <- subset(angio_data_long, TreeName == "12907*N")

ov6 <- subset(angio_data_long, TreeName == "12907*Q")

ov7 <- subset(angio_data_long, TreeName == "6986*A")

ov8 <- subset(angio_data_long, TreeName == "7321*A")

ov9 <- subset(angio_data_long, TreeName == "7947*A")

ov10 <- subset(angio_data_long, TreeName == "carova1")

ov11 <- subset(angio_data_long, TreeName == "carova2")

ov12 <- subset(angio_data_long, TreeName == "carova3")


plot(ov1$Year, y = ov1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab = "R
lines(ov2$Year, y = ov2$Ringwidth, type = "o", col = "blue")
lines(ov3$Year, y = ov3$Ringwidth, type = "o", col = "purple")
lines(ov4$Year, y = ov4$Ringwidth, type = "o", col = "lightblue")
lines(ov5$Year, y = ov5$Ringwidth, type = "o", col = "lightgreen")
lines(ov6$Year, y = ov6$Ringwidth, type = "o", col = "pink")
lines(ov7$Year, y = ov7$Ringwidth, type = "o", col = "red")
lines(ov8$Year, y = ov8$Ringwidth, type = "o", col = "orange")
lines(ov9$Year, y = ov9$Ringwidth, type = "o", col = "yellow")
lines(ov10$Year, y = ov10$Ringwidth, type = "o", col = "deeppink")
lines(ov11$Year, y = ov11$Ringwidth, type = "o", col = "burlywood")
lines(ov12$Year, y = ov12$Ringwidth, type = "o", col = "black")

legend("topright", inset = .05, title = "Tree name", c("12907*B", "12907*D", "12907*G", "12907*I"
```



Catalpa speciosa

```
speci1 <- subset(angio_data_long, TreeName == "1245-79*A")

speci2 <- subset(angio_data_long, TreeName == "1245-79*B")

speci3 <- subset(angio_data_long, TreeName == "catspe1")

speci4 <- subset(angio_data_long, TreeName == "catspe2")

speci5 <- subset(angio_data_long, TreeName == "catspe3")

speci6 <- subset(angio_data_long, TreeName == "catspe4")


plot(speci1$Year, y = speci1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(speci2$Year, y = speci2$Ringwidth, type = "o", col = "blue")
lines(speci3$Year, y = speci3$Ringwidth, type = "o", col = "purple")
lines(speci4$Year, y = speci4$Ringwidth, type = "o", col = "lightgreen")
lines(speci5$Year, y = speci5$Ringwidth, type = "o", col = "pink")
lines(speci6$Year, y = speci6$Ringwidth, type = "o", col = "deeppink")
```

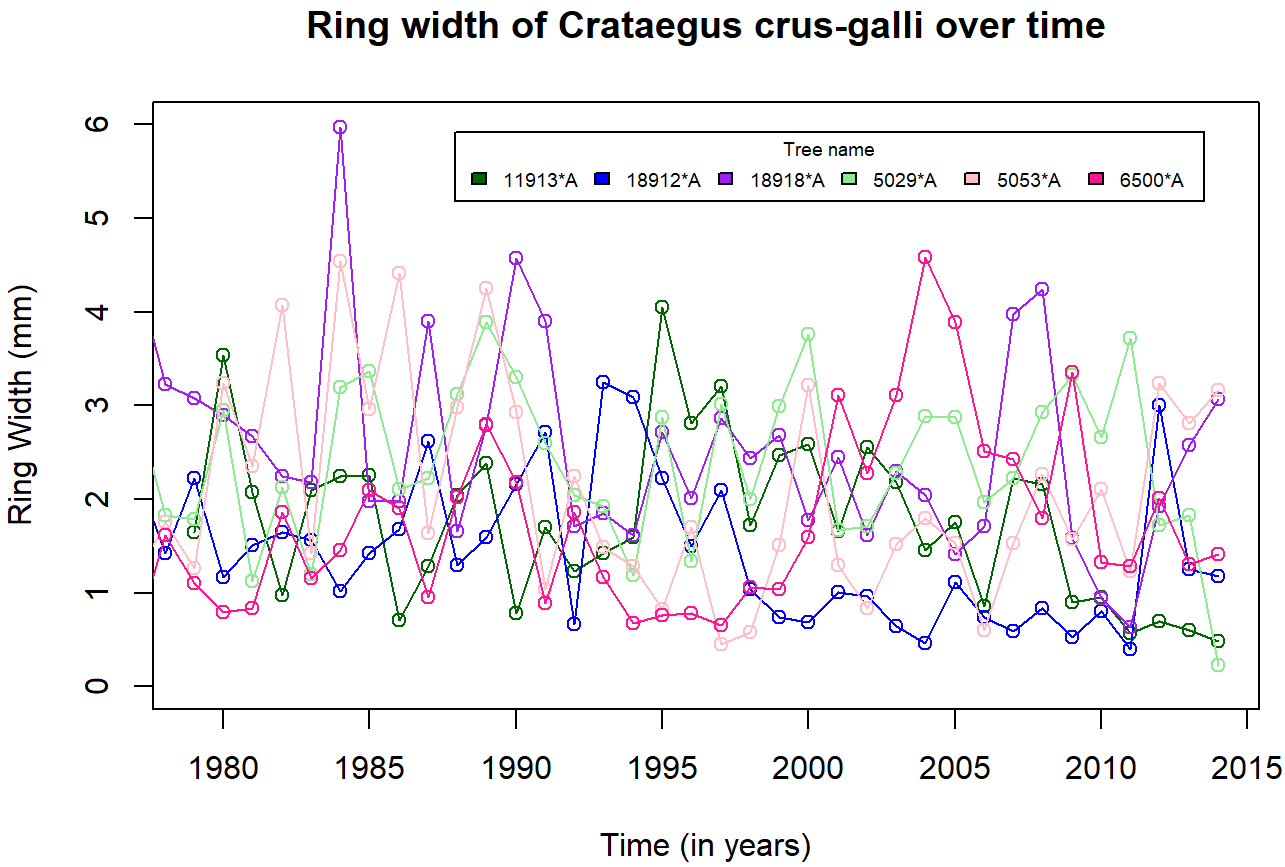
```
legend("top", inset = .05, title = "Tree name", c("1245-79*A", "1245-79*B", "catspe1", "catspe2",
```



Crataegus crus-galli

```
crga1 <- subset(angio_data_long, TreeName == "11913*A")

crga2 <- subset(angio_data_long, TreeName == "18912*A")

crga3 <- subset(angio_data_long, TreeName == "18918*A")

crga4 <- subset(angio_data_long, TreeName == "5029*A")

crga5 <- subset(angio_data_long, TreeName == "5053*A")

crga6 <- subset(angio_data_long, TreeName == "6500*A")


plot(crga1$Year, y = crga1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(crga2$Year, y = crga2$Ringwidth, type = "o", col = "blue")
lines(crga3$Year, y = crga3$Ringwidth, type = "o", col = "purple")
lines(crga4$Year, y = crga4$Ringwidth, type = "o", col = "lightgreen")
```

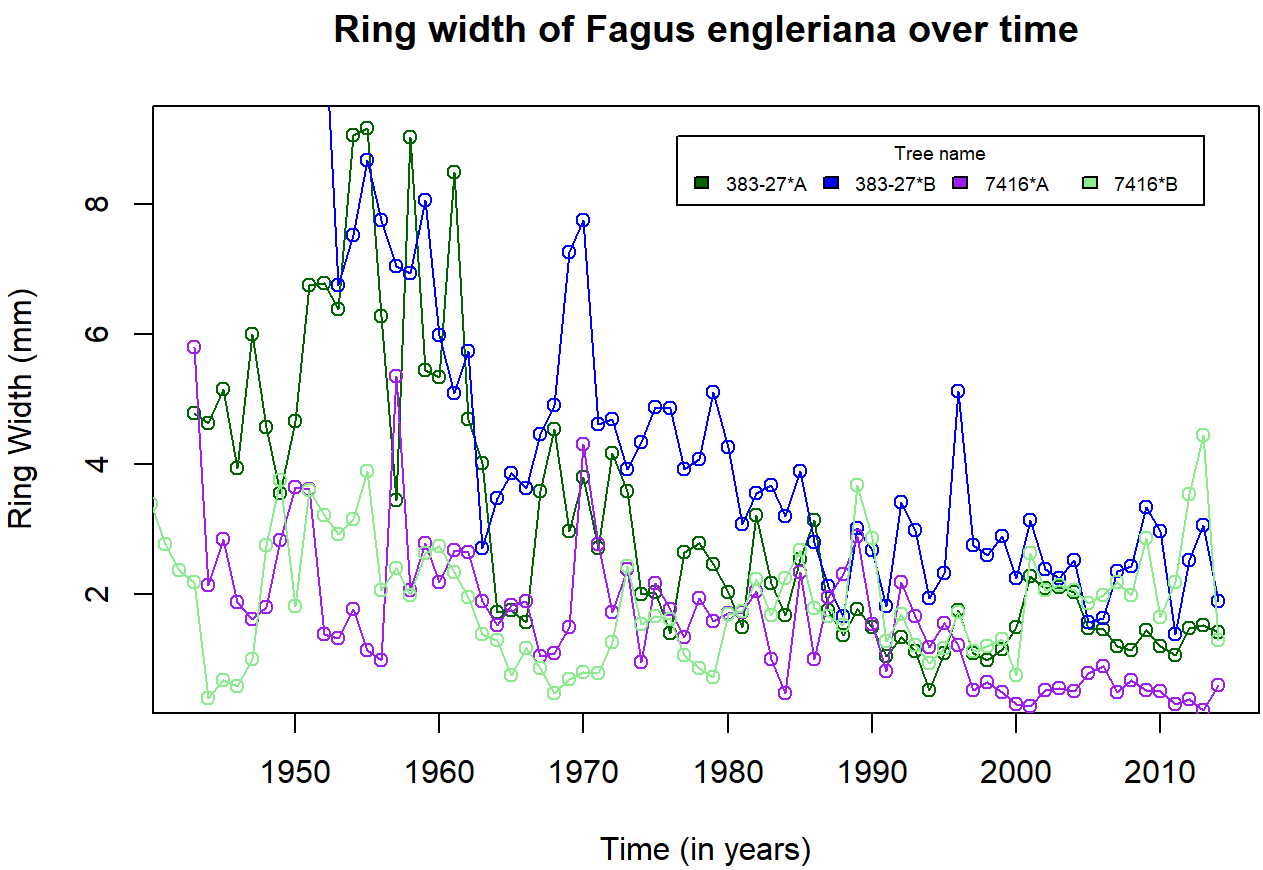
```
lines(crga5$Year, y = crga5$Ringwidth, type = "o", col = "pink")
lines(crga6$Year, y = crga6$Ringwidth, type = "o", col = "deeppink")

legend("topright", inset = .05, title = "Tree name", c("11913*A", "18912*A", "18918*A", "5029*A",
```



Fagus engleriana

```
engl1 <- subset(angio_data_long, TreeName == "383-27*A")

engl2 <- subset(angio_data_long, TreeName == "383-27*B")

engl3 <- subset(angio_data_long, TreeName == "7416*A")

engl4 <- subset(angio_data_long, TreeName == "7416*B")


plot(engl1$Year, y = engl1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(engl2$Year, y = engl2$Ringwidth, type = "o", col = "blue")
lines(engl3$Year, y = engl3$Ringwidth, type = "o", col = "purple")
lines(engl4$Year, y = engl4$Ringwidth, type = "o", col = "lightgreen")
```

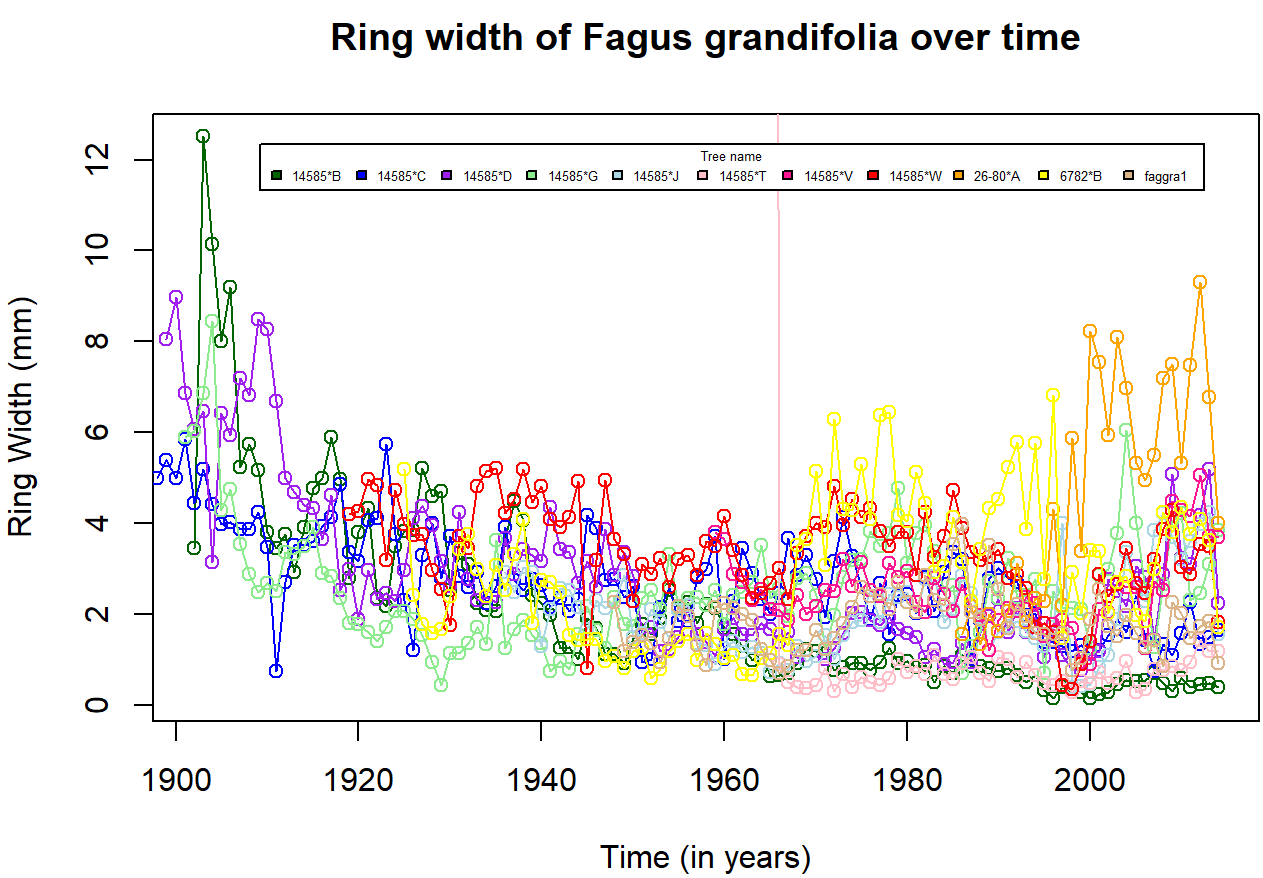
```
legend("topright", inset = .05, title = "Tree name", c("383-27*A", "383-27*B", "7416*A", "7416*B"
```



Fagus grandifolia

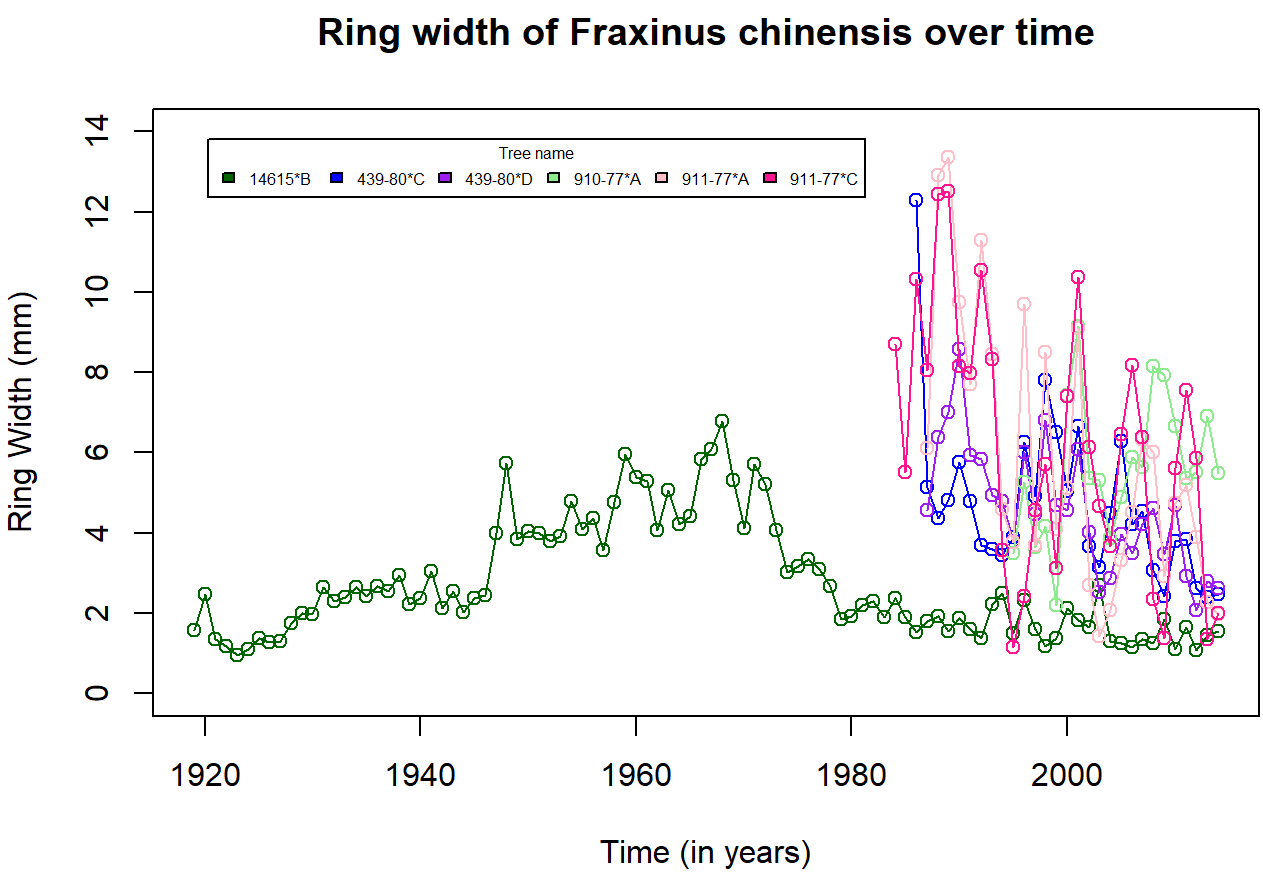
```
grandi1 <- subset(angio_data_long, TreeName == "14585*B")

grandi2 <- subset(angio_data_long, TreeName == "14585*C")

grandi3 <- subset(angio_data_long, TreeName == "14585*D")

grandi4 <- subset(angio_data_long, TreeName == "14585*G")

grandi5 <- subset(angio_data_long, TreeName == "14585*J")

grandi6 <- subset(angio_data_long, TreeName == "14585*T")

grandi7 <- subset(angio_data_long, TreeName == "14585*V")

grandi8 <- subset(angio_data_long, TreeName == "14585*W")

grandi9 <- subset(angio_data_long, TreeName == "26-80*A")
```

```
grandi10 <- subset(angio_data_long, TreeName == "6782*B")

grandi11 <- subset(angio_data_long, TreeName == "faggra1")


plot(grandi1$Year, y = grandi1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", y
lines(grandi2$Year, y = grandi2$Ringwidth, type = "o", col = "blue")
lines(grandi3$Year, y = grandi3$Ringwidth, type = "o", col = "purple")
lines(grandi4$Year, y = grandi4$Ringwidth, type = "o", col = "lightgreen")
lines(grandi5$Year, y = grandi5$Ringwidth, type = "o", col = "lightblue")
lines(grandi6$Year, y = grandi6$Ringwidth, type = "o", col = "pink")
lines(grandi7$Year, y = grandi7$Ringwidth, type = "o", col = "deeppink")
lines(grandi8$Year, y = grandi8$Ringwidth, type = "o", col = "red")
lines(grandi9$Year, y = grandi9$Ringwidth, type = "o", col = "orange")
lines(grandi10$Year, y = grandi10$Ringwidth, type = "o", col = "yellow")
lines(grandi11$Year, y = grandi11$Ringwidth, type = "o", col = "burlywood")

legend("topright", inset = .05, title = "Tree name", c("14585*B", "14585*C", "14585*D", "14585*G"
```



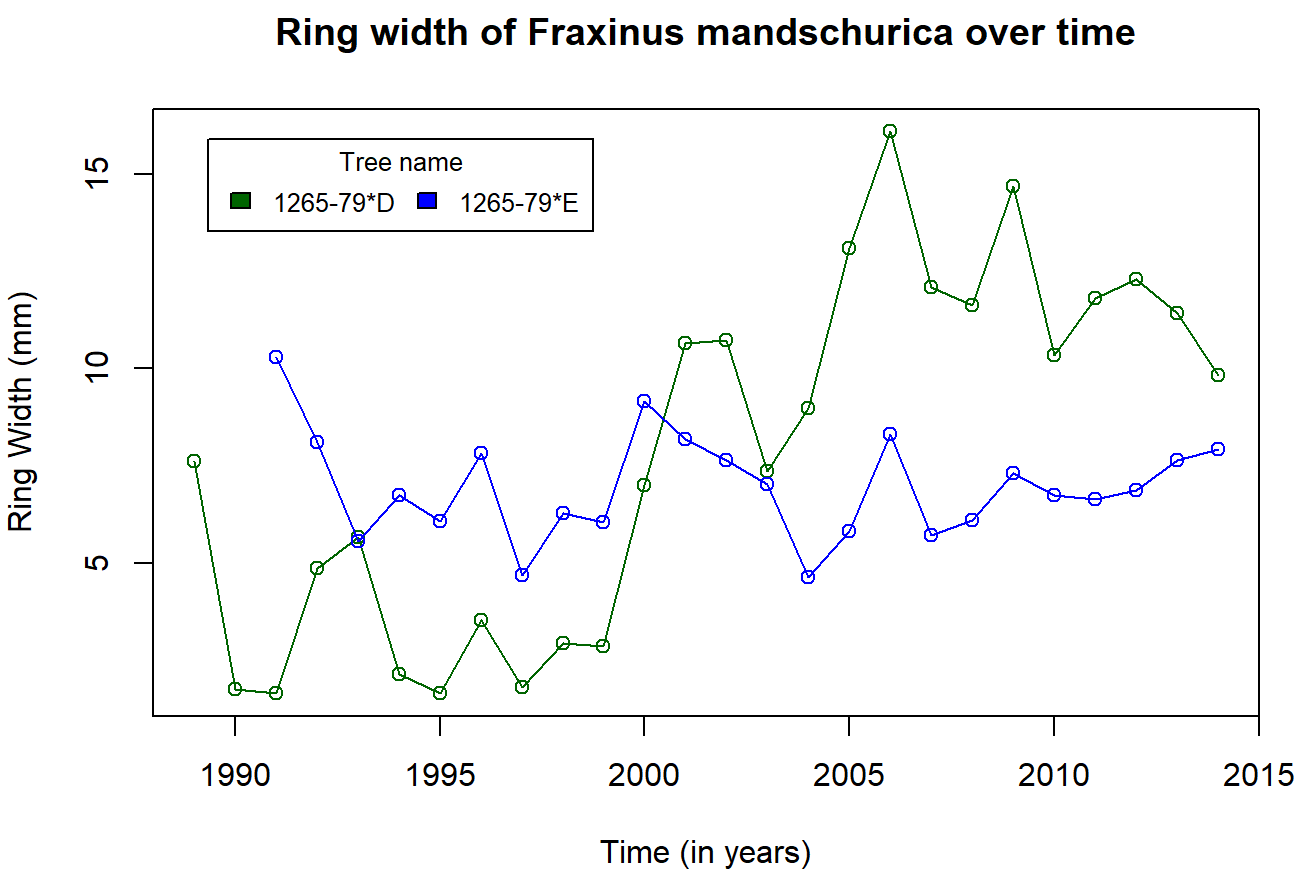
Fraxinus chinensis

```
chin1 <- subset(angio_data_long, TreeName == "14615*B")

chin2 <- subset(angio_data_long, TreeName == "439-80*C")

chin3 <- subset(angio_data_long, TreeName == "439-80*D")

chin4 <- subset(angio_data_long, TreeName == "910-77*A")

chin5 <- subset(angio_data_long, TreeName == "911-77*A")

chin6 <- subset(angio_data_long, TreeName == "911-77*C")


plot(chin1$Year, y = chin1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(chin2$Year, y = chin2$Ringwidth, type = "o", col = "blue")
lines(chin3$Year, y = chin3$Ringwidth, type = "o", col = "purple")
lines(chin4$Year, y = chin4$Ringwidth, type = "o", col = "lightgreen")
lines(chin5$Year, y = chin5$Ringwidth, type = "o", col = "pink")
lines(chin6$Year, y = chin6$Ringwidth, type = "o", col = "deeppink")

legend("topleft", inset = .05, title = "Tree name", c("14615*B", "439-80*C", "439-80*D", "910-77*A
```



Fraxinus mandschurica

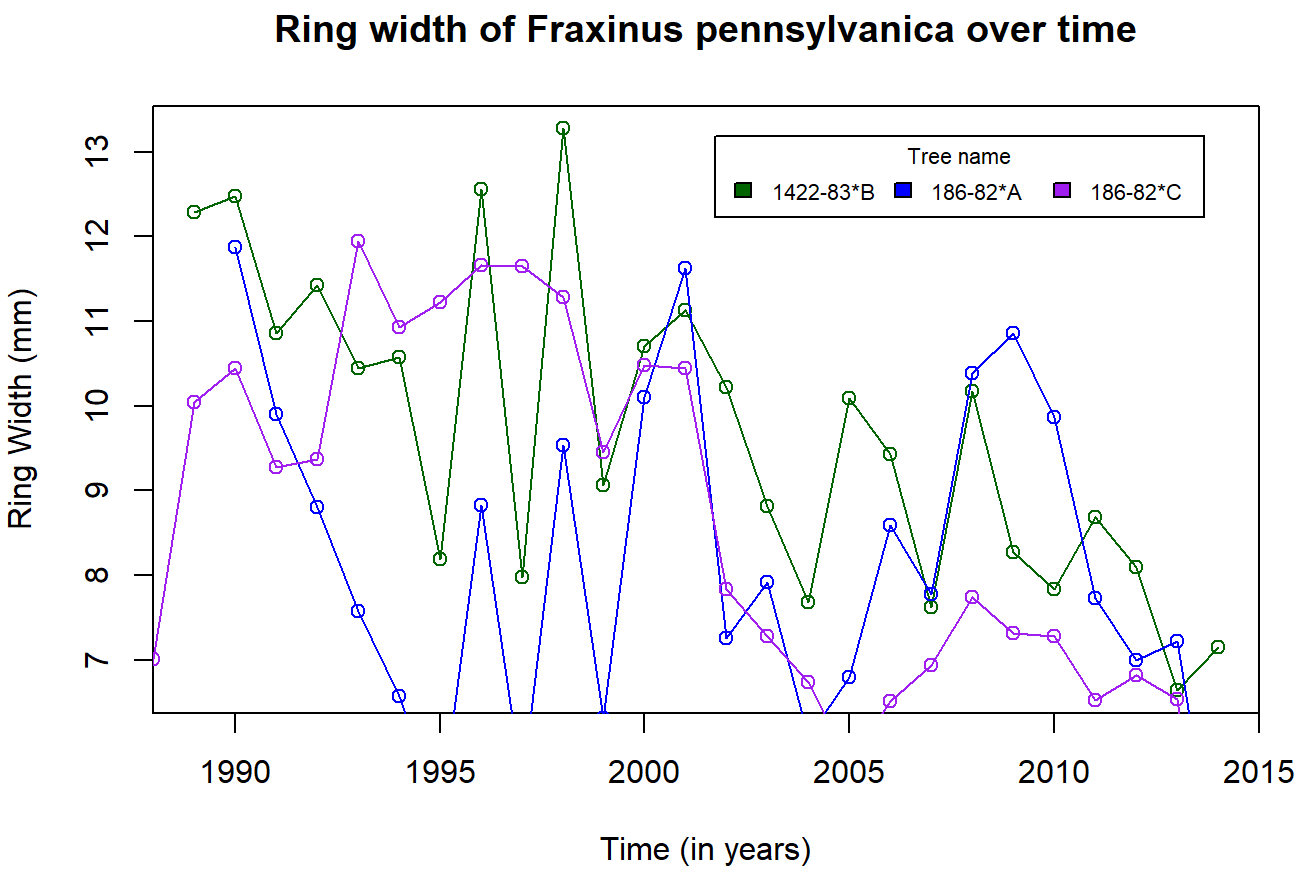
```
mands1 <- subset(angio_data_long, TreeName == "1265-79*D")

mands2 <- subset(angio_data_long, TreeName == "1265-79*E")


plot(mands1$Year, y = mands1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(mands2$Year, y = mands2$Ringwidth, type = "o", col = "blue")

legend("topleft", inset = .05, title = "Tree name", c("1265-79*D", "1265-79*E"), fill = c("darkgr
```



Fraxinus pennsylvanica

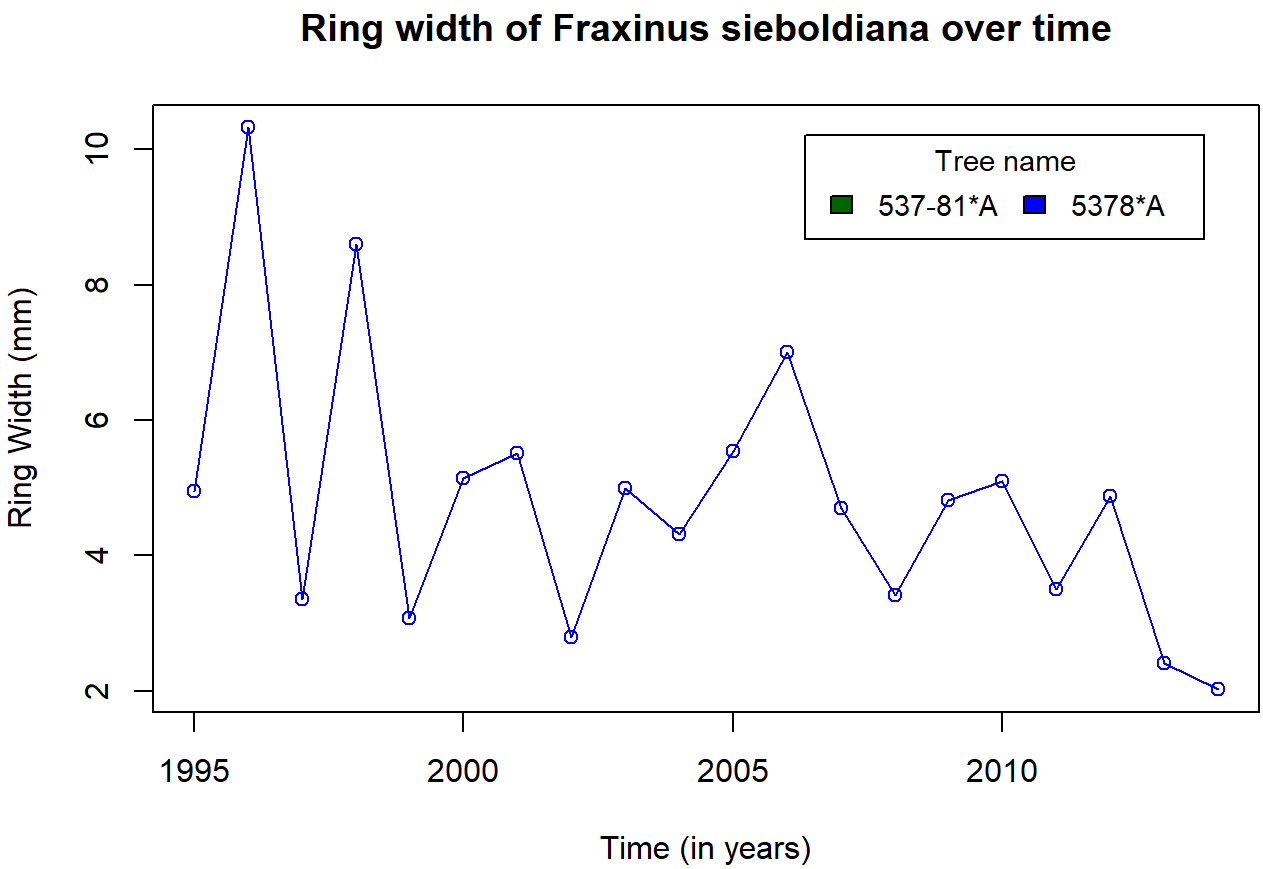
```
penn1 <- subset(angio_data_long, TreeName == "1422-83*B")

penn2 <- subset(angio_data_long, TreeName == "186-82*A")

penn3 <- subset(angio_data_long, TreeName == "186-82*C")
```

```
plot(penn1$Year, y = penn1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(penn2$Year, y = penn2$Ringwidth, type = "o", col = "blue")
lines(penn3$Year, y = penn3$Ringwidth, type = "o", col = "purple")

legend("topright", inset = .05, title = "Tree name", c("1422-83*B", "186-82*A", "186-82*C"), fill
```



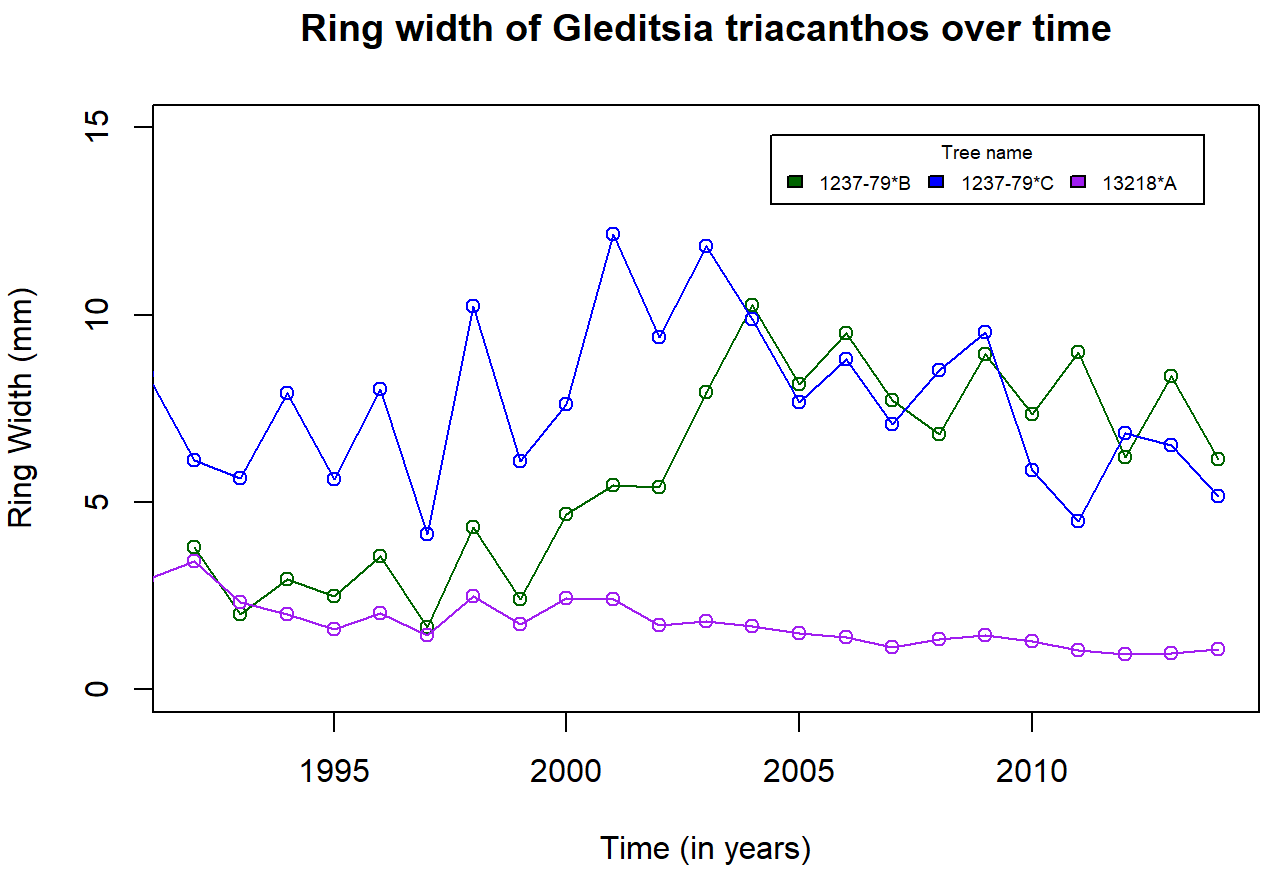
Fraxinus sieboldiana

```
sieb1 <- subset(angio_data_long, TreeName == "537-81*A")

sieb2 <- subset(angio_data_long, TreeName == "5378*A")


plot(sieb1$Year, y = sieb1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(sieb2$Year, y = sieb2$Ringwidth, type = "o", col = "blue")

legend("topright", inset = .05, title = "Tree name", c("537-81*A", "5378*A"), fill = c("darkgreen
```



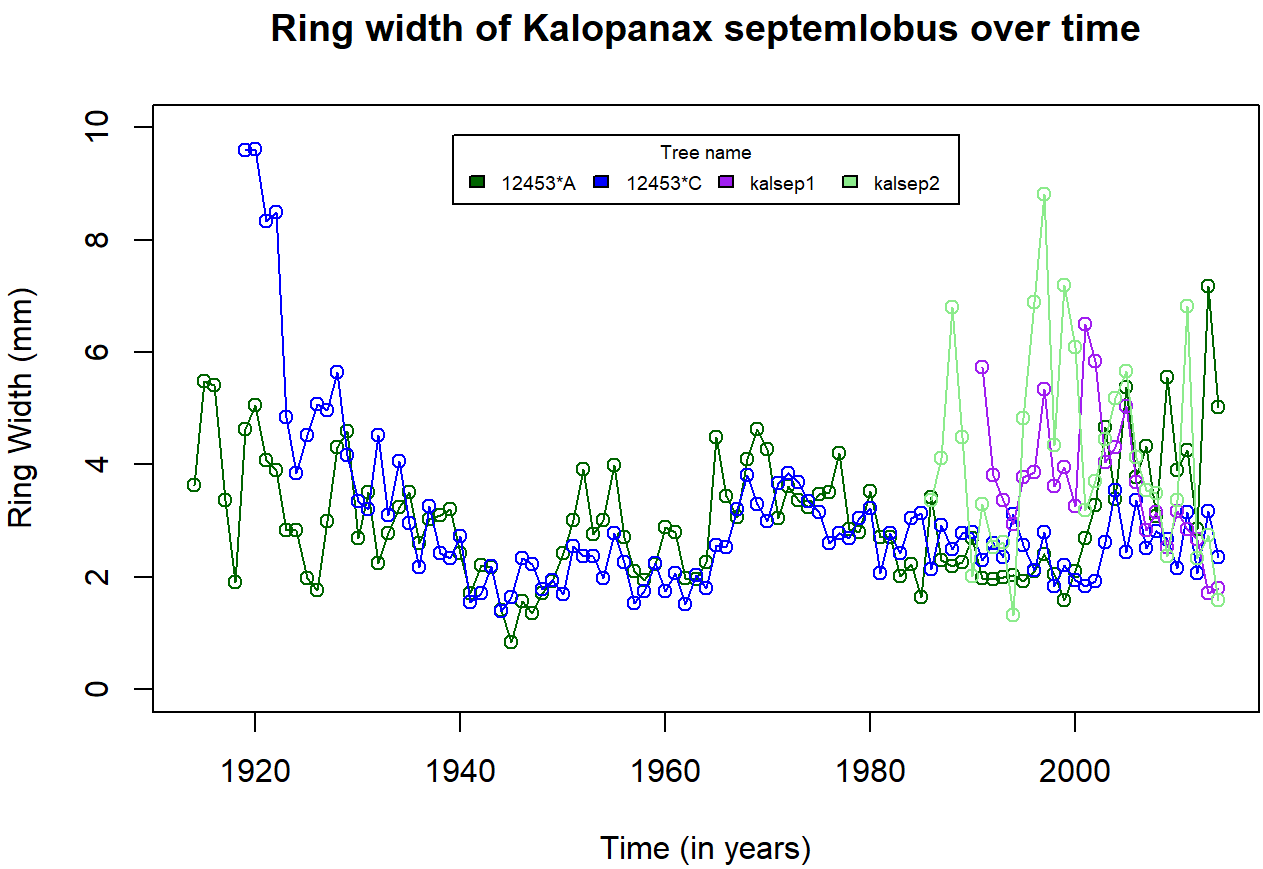
Gleditsia triacanthos

```
tria1 <- subset(angio_data_long, TreeName == "1237-79*B")

tria2 <- subset(angio_data_long, TreeName == "1237-79*C")

tria3 <- subset(angio_data_long, TreeName == "13218*A")


plot(tria1$Year, y = tria1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(tria2$Year, y = tria2$Ringwidth, type = "o", col = "blue")
lines(tria3$Year, y = tria3$Ringwidth, type = "o", col = "purple")

legend("topright", inset = .05, title = "Tree name", c("1237-79*B", "1237-79*C", "13218*A"), fill
```



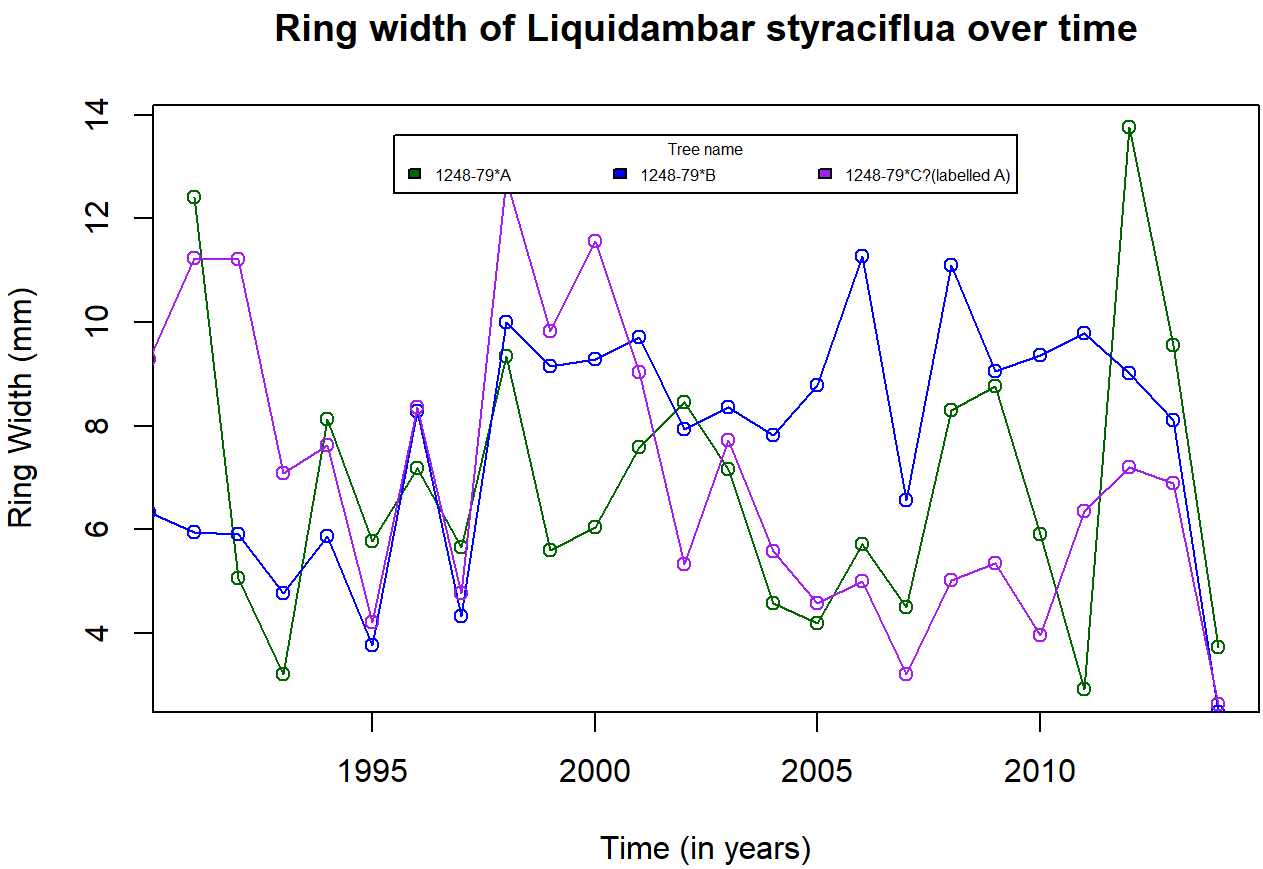
Kalopanax septemlobus

```
sept1 <- subset(angio_data_long, TreeName == "12453*A")

sept2 <- subset(angio_data_long, TreeName == "12453*C")

sept3 <- subset(angio_data_long, TreeName == "kalsep1")

sept4 <- subset(angio_data_long, TreeName == "kalsep2")


plot(sept1$Year, y = sept1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(sept2$Year, y = sept2$Ringwidth, type = "o", col = "blue")
lines(sept3$Year, y = sept3$Ringwidth, type = "o", col = "purple")
lines(sept4$Year, y = sept4$Ringwidth, type = "o", col = "lightgreen")

legend("top", inset = .05, title = "Tree name", c("12453*A", "12453*C", "kalsep1", "kalsep2"), fi
```



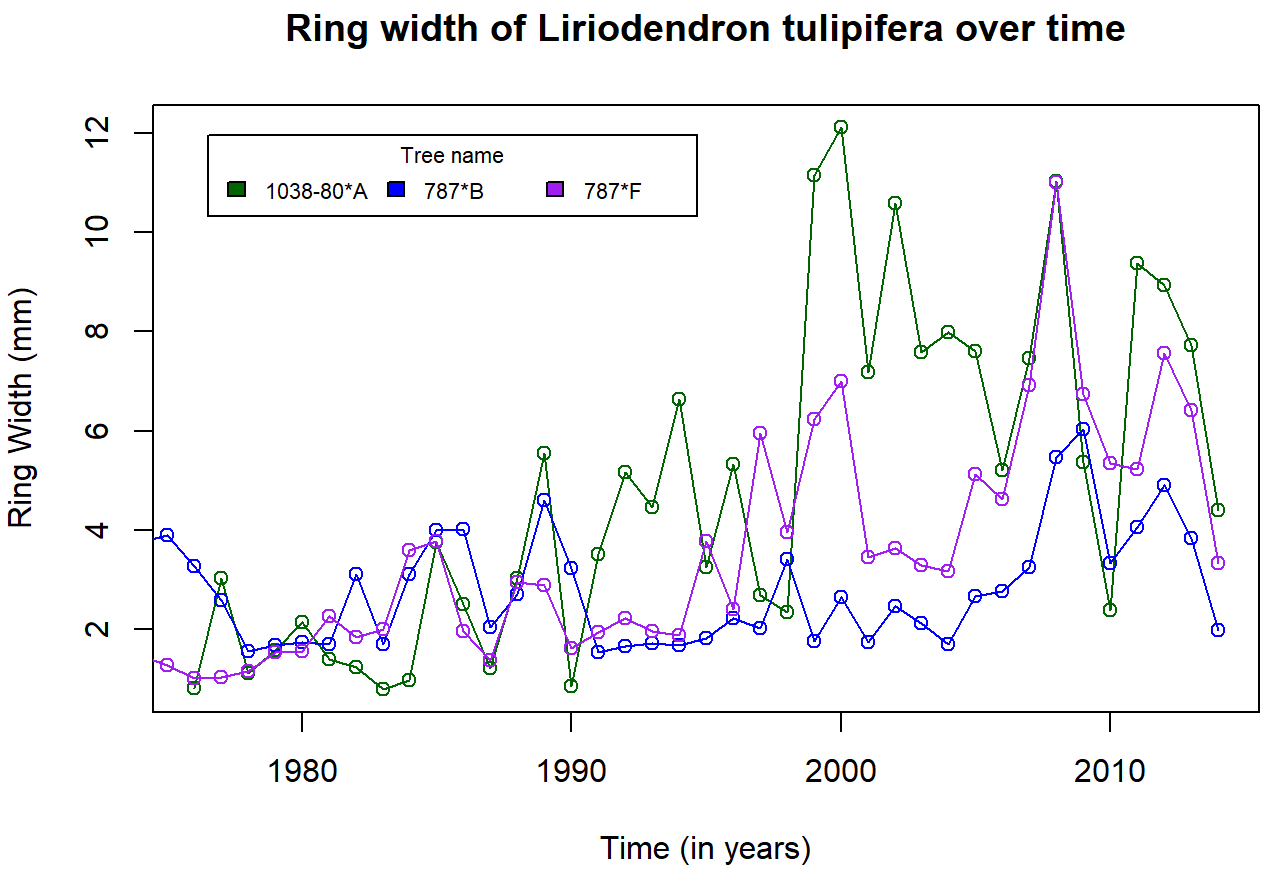
Liquidambar styraciflua

```
styra1 <- subset(angio_data_long, TreeName == "1248-79*A")

styra2 <- subset(angio_data_long, TreeName == "1248-79*B")

styra3 <- subset(angio_data_long, TreeName == "1248-79*C?(labelled A)")


plot(styra1$Year, y = styra1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(styra2$Year, y = styra2$Ringwidth, type = "o", col = "blue")
lines(styra3$Year, y = styra3$Ringwidth, type = "o", col = "purple")

legend("top", inset = .05, title = "Tree name", c("1248-79*A", "1248-79*B", "1248-79*C?(labelled A
```



Liriodendron tulipifera

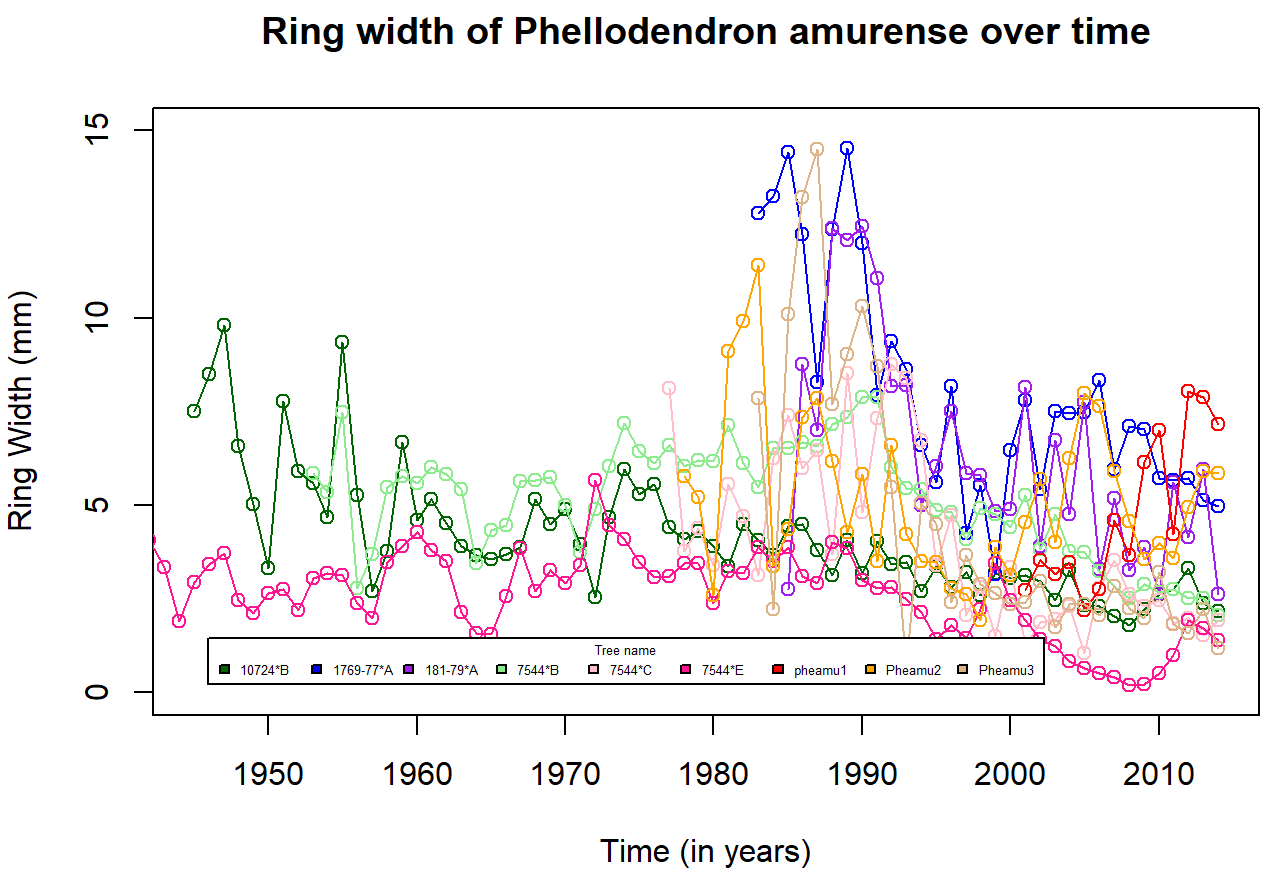
```
tulip1 <- subset(angio_data_long, TreeName == "1038-80*A")

tulip2 <- subset(angio_data_long, TreeName == "787*B")

tulip3 <- subset(angio_data_long, TreeName == "787*F")


plot(tulip1$Year, y = tulip1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(tulip2$Year, y = tulip2$Ringwidth, type = "o", col = "blue")
lines(tulip3$Year, y = tulip3$Ringwidth, type = "o", col = "purple")

legend("topleft", inset = .05, title = "Tree name", c("1038-80*A", "787*B", "787*F"), fill = c("da
```



Phellodendron amurense

```
amur1 <- subset(angio_data_long, TreeName == "10724*B")

amur2 <- subset(angio_data_long, TreeName == "1769-77*A")

amur3 <- subset(angio_data_long, TreeName == "181-79*A")

amur4 <- subset(angio_data_long, TreeName == "7544*B")

amur5 <- subset(angio_data_long, TreeName == "7544*C")

amur6 <- subset(angio_data_long, TreeName == "7544*E")

amur7 <- subset(angio_data_long, TreeName == "pheamu1")

amur8 <- subset(angio_data_long, TreeName == "Pheamu2")

amur9 <- subset(angio_data_long, TreeName == "Pheamu3")


plot(amur1$Year, y = amur1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(amur2$Year, y = amur2$Ringwidth, type = "o", col = "blue")
```

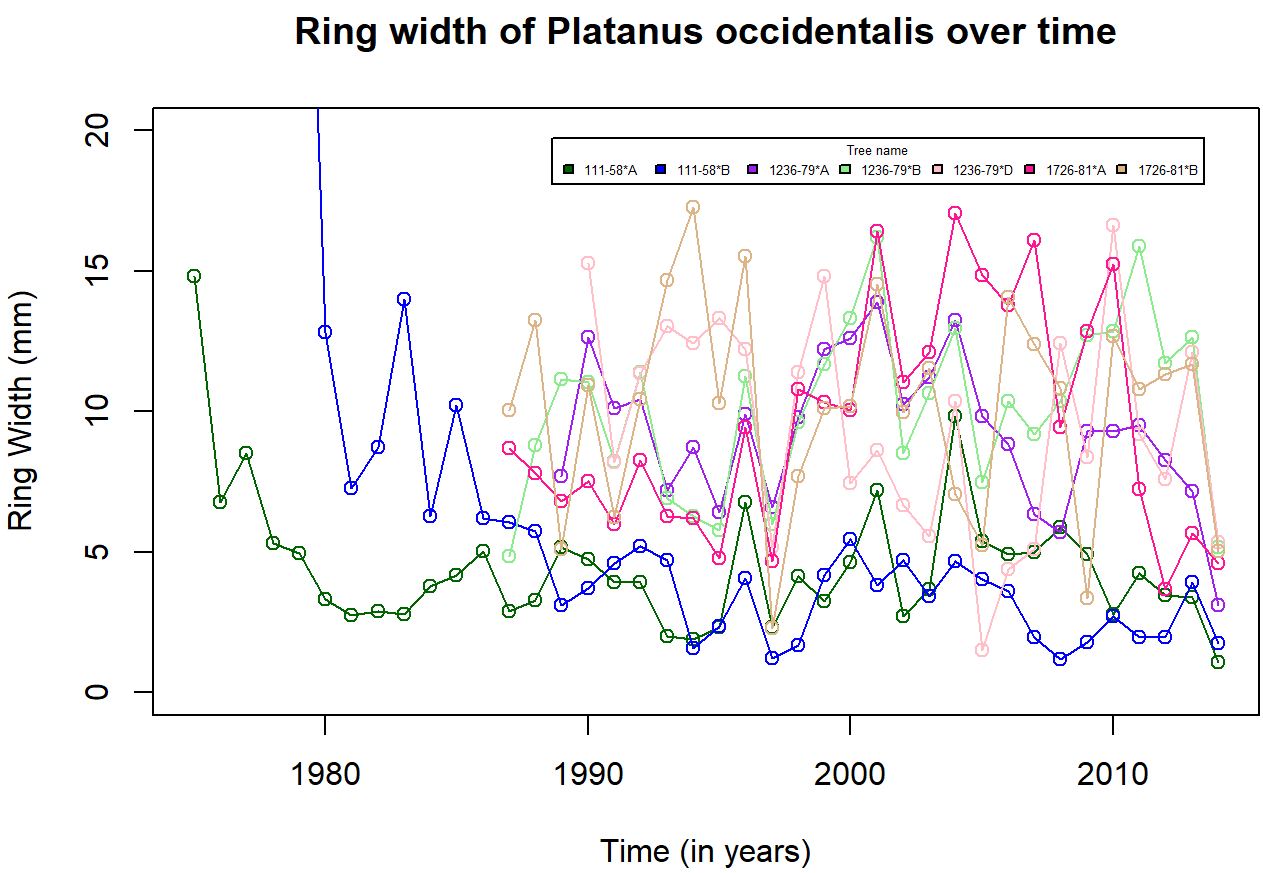
```
lines(amur3$Year, y = amur3$Ringwidth, type = "o", col = "purple")
lines(amur4$Year, y = amur4$Ringwidth, type = "o", col = "lightgreen")
lines(amur5$Year, y = amur5$Ringwidth, type = "o", col = "pink")
lines(amur6$Year, y = amur6$Ringwidth, type = "o", col = "deeppink")
lines(amur7$Year, y = amur7$Ringwidth, type = "o", col = "red")
lines(amur8$Year, y = amur8$Ringwidth, type = "o", col = "orange")
lines(amur9$Year, y = amur9$Ringwidth, type = "o", col = "burlywood")

legend("bottomleft", inset = .05, title = "Tree name", c("10724*B", "1769-77*A", "181-79*A", "754
```



Platanus occidentalis

```
occi1 <- subset(angio_data_long, TreeName == "111-58*A")

occi2 <- subset(angio_data_long, TreeName == "111-58*B")

occi3 <- subset(angio_data_long, TreeName == "1236-79*A")

occi4 <- subset(angio_data_long, TreeName == "1236-79*B")

occi5 <- subset(angio_data_long, TreeName == "1236-79*D")
```

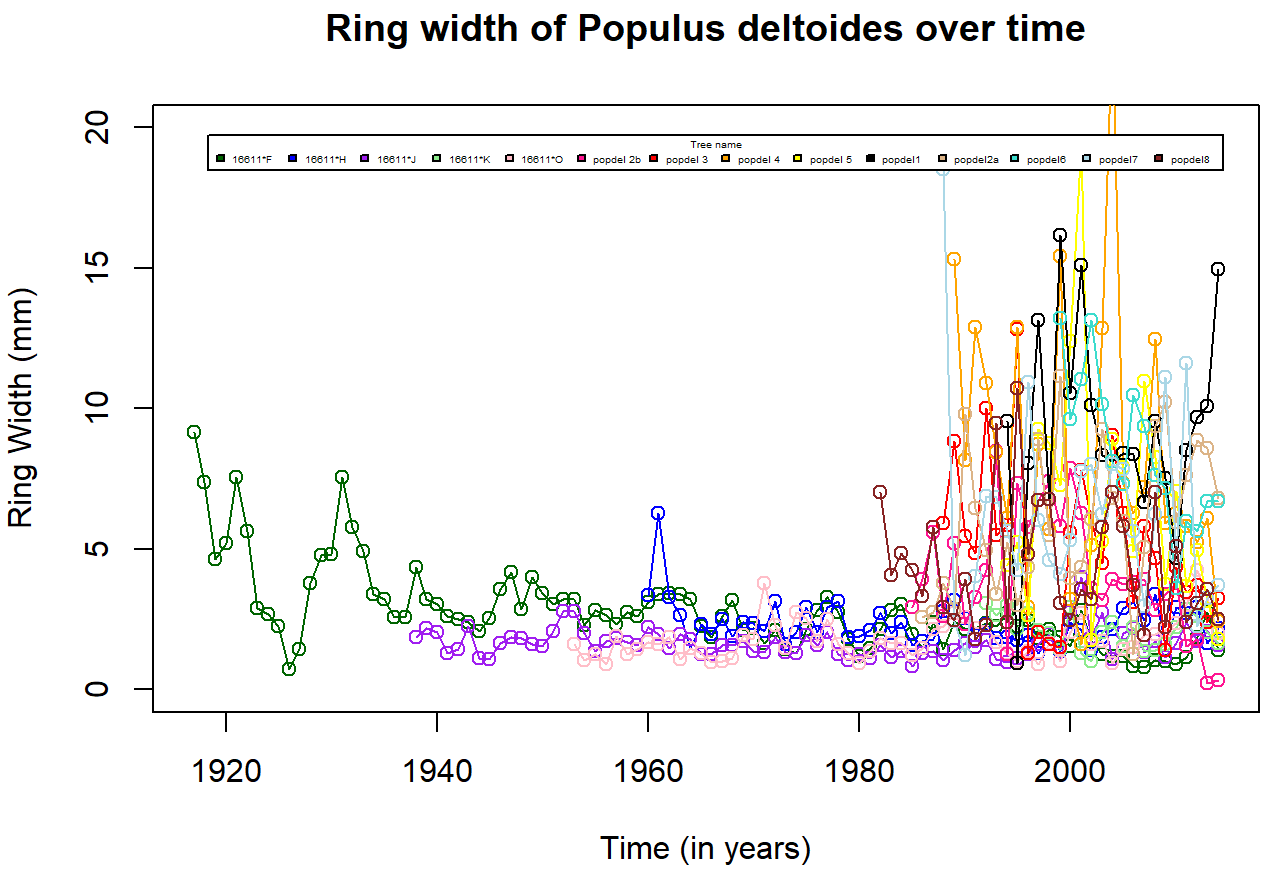
```
occi6 <- subset(angio_data_long, TreeName == "1726-81*A")

occi7 <- subset(angio_data_long, TreeName == "1726-81*B")


plot(occi1$Year, y = occi1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(occi2$Year, y = occi2$Ringwidth, type = "o", col = "blue")
lines(occi3$Year, y = occi3$Ringwidth, type = "o", col = "purple")
lines(occi4$Year, y = occi4$Ringwidth, type = "o", col = "lightgreen")
lines(occi5$Year, y = occi5$Ringwidth, type = "o", col = "pink")
lines(occi6$Year, y = occi6$Ringwidth, type = "o", col = "deeppink")
lines(occi7$Year, y = occi7$Ringwidth, type = "o", col = "burlywood")

legend("topright", inset = .05, title = "Tree name", c("111-58*A", "111-58*B", "1236-79*A", "1236
```



Populus deltoides

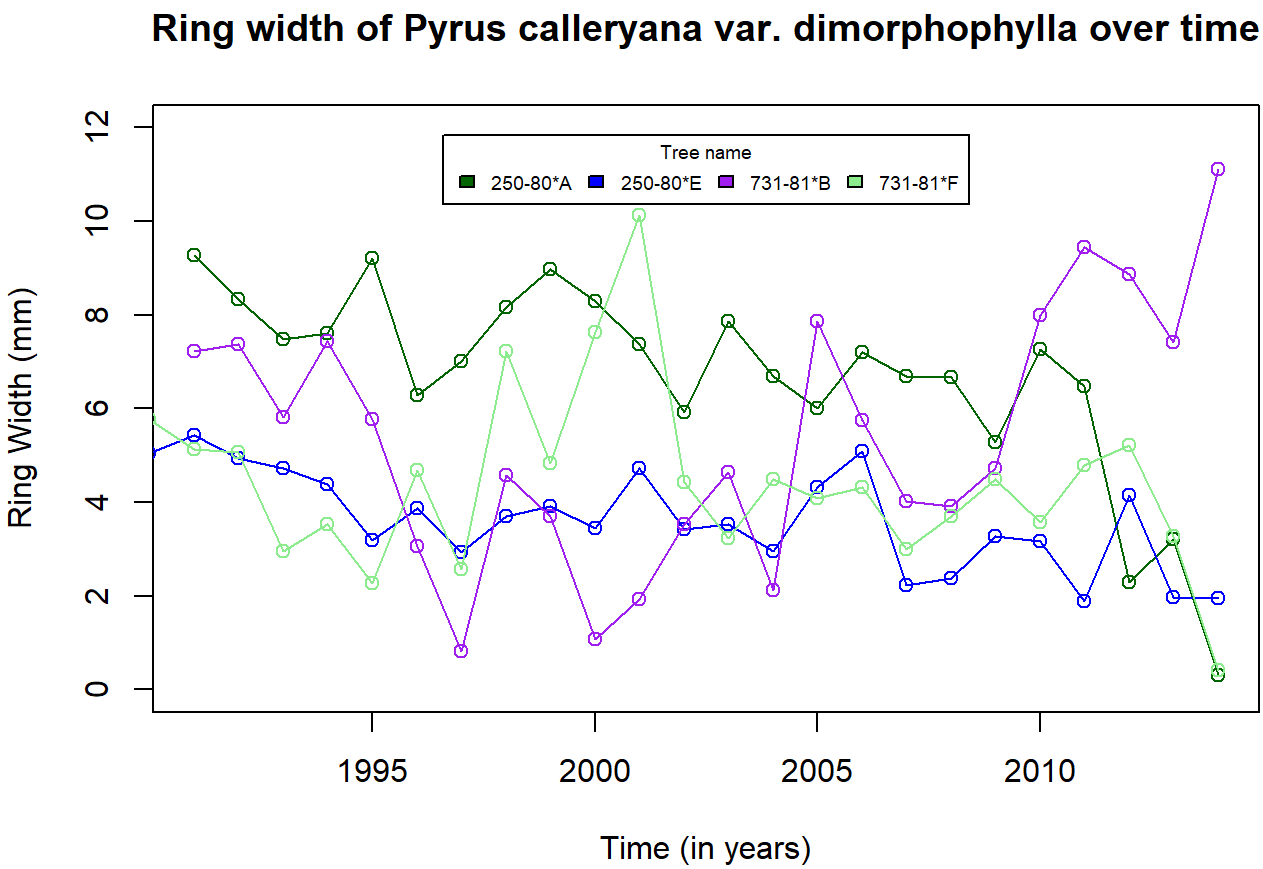
```
delto1 <- subset(angio_data_long, TreeName == "16611*F")

delto2 <- subset(angio_data_long, TreeName == "16611*H")

delto3 <- subset(angio_data_long, TreeName == "16611*J")
```

```
delto4 <- subset(angio_data_long, TreeName == "16611*K")

delto5 <- subset(angio_data_long, TreeName == "16611*O")

delto6 <- subset(angio_data_long, TreeName == "popdel 2b")

delto7 <- subset(angio_data_long, TreeName == "popdel 3")

delto8 <- subset(angio_data_long, TreeName == "popdel 4")

delto9 <- subset(angio_data_long, TreeName == "popdel 5")

delto10 <- subset(angio_data_long, TreeName == "popdel1")

delto11 <- subset(angio_data_long, TreeName == "popdel2a")

delto12 <- subset(angio_data_long, TreeName == "popdel6")

delto13 <- subset(angio_data_long, TreeName == "popdel7")

delto14 <- subset(angio_data_long, TreeName == "popdel8")


plot(delto1$Year, y = delto1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(delto2$Year, y = delto2$Ringwidth, type = "o", col = "blue")
lines(delto3$Year, y = delto3$Ringwidth, type = "o", col = "purple")
lines(delto4$Year, y = delto4$Ringwidth, type = "o", col = "lightgreen")
lines(delto5$Year, y = delto5$Ringwidth, type = "o", col = "pink")
lines(delto6$Year, y = delto6$Ringwidth, type = "o", col = "deeppink")
lines(delto7$Year, y = delto7$Ringwidth, type = "o", col = "red")
lines(delto8$Year, y = delto8$Ringwidth, type = "o", col = "orange")
lines(delto9$Year, y = delto9$Ringwidth, type = "o", col = "yellow")
lines(delto10$Year, y = delto10$Ringwidth, type = "o", col = "black")
lines(delto11$Year, y = delto11$Ringwidth, type = "o", col = "burlywood")
lines(delto12$Year, y = delto12$Ringwidth, type = "o", col = "turquoise")
lines(delto13$Year, y = delto13$Ringwidth, type = "o", col = "lightblue")
lines(delto14$Year, y = delto14$Ringwidth, type = "o", col = "brown4")

legend("topleft", inset = .05, title = "Tree name", c("16611*F", "16611*H", "16611*J", "16611*K",
```



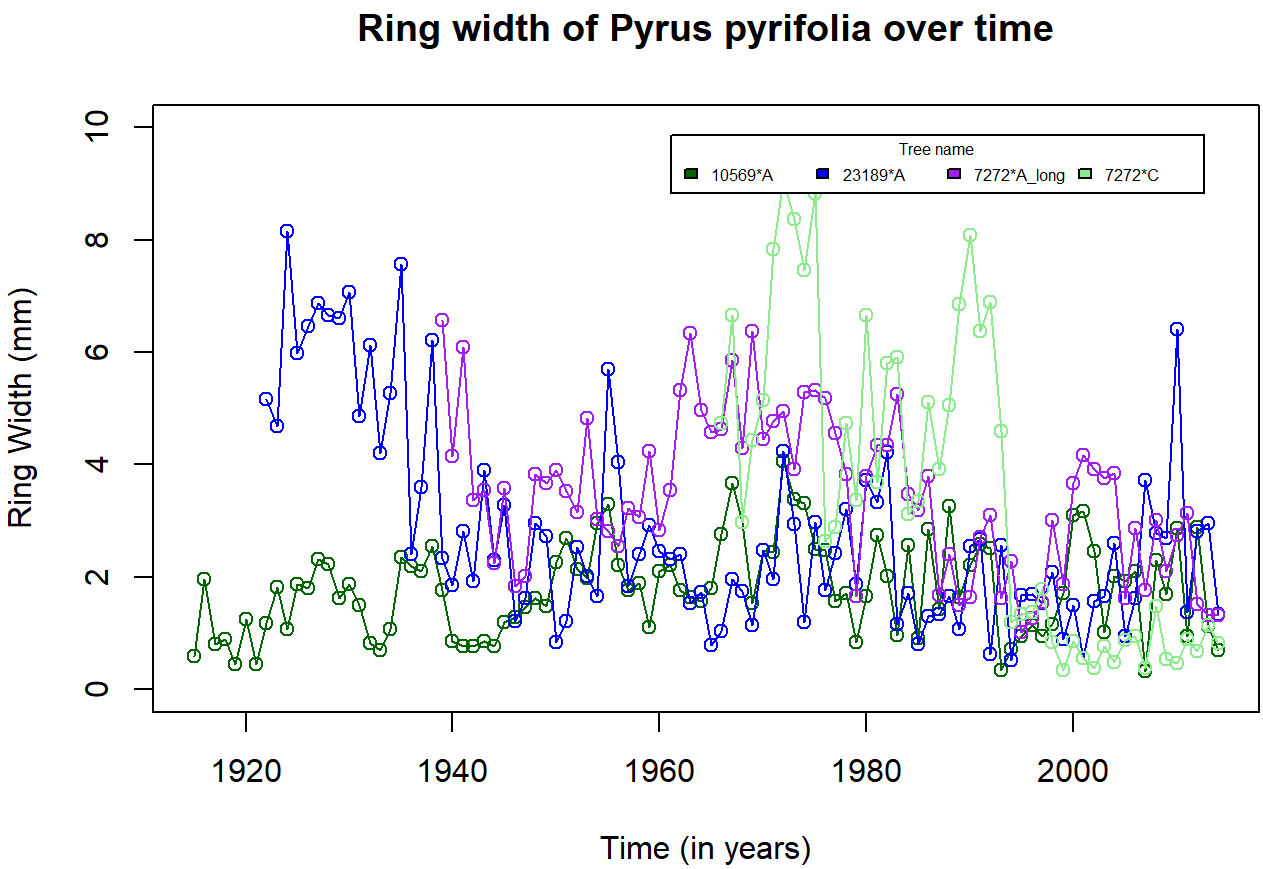
Pyrus calleryana var. dimorphophylla

```
calle1 <- subset(angio_data_long, TreeName == "250-80*A")

calle2 <- subset(angio_data_long, TreeName == "250-80*E")

calle3 <- subset(angio_data_long, TreeName == "731-81*B")

calle4 <- subset(angio_data_long, TreeName == "731-81*F")


plot(calle1$Year, y = calle1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(calle2$Year, y = calle2$Ringwidth, type = "o", col = "blue")
lines(calle3$Year, y = calle3$Ringwidth, type = "o", col = "purple")
lines(calle4$Year, y = calle4$Ringwidth, type = "o", col = "lightgreen")

legend("top", inset = .05, title = "Tree name", c("250-80*A", "250-80*E", "731-81*B", "731-81*F")
```



Pyrus pyrifolia

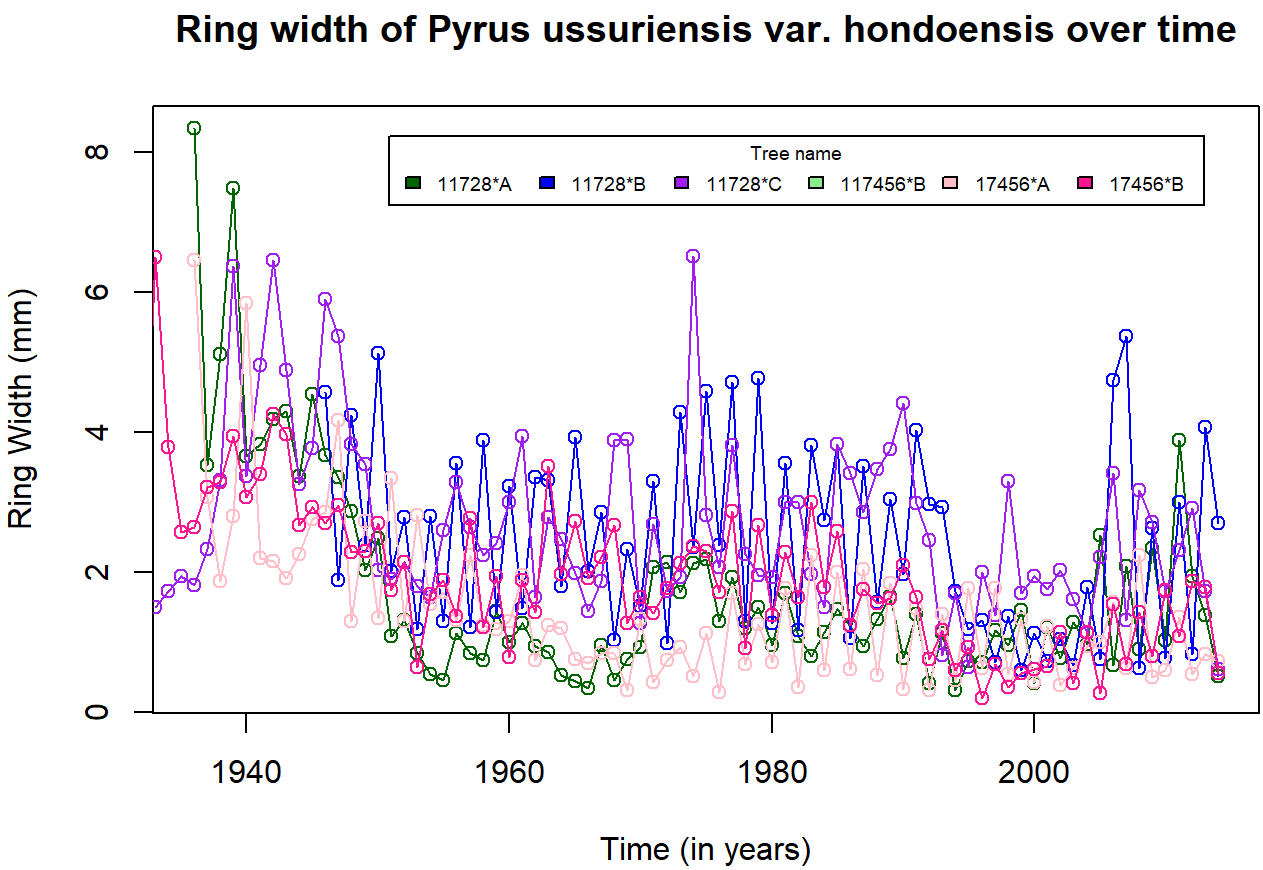
```
pyri1 <- subset(angio_data_long, TreeName == "10569*A")

pyri2 <- subset(angio_data_long, TreeName == "23189*A")

pyri3 <- subset(angio_data_long, TreeName == "7272*A_long")

pyri4 <- subset(angio_data_long, TreeName == "7272*C")


plot(pyri1$Year, y = pyri1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(pyri2$Year, y = pyri2$Ringwidth, type = "o", col = "blue")
lines(pyri3$Year, y = pyri3$Ringwidth, type = "o", col = "purple")
lines(pyri4$Year, y = pyri4$Ringwidth, type = "o", col = "lightgreen")

legend("topright", inset = .05, title = "Tree name", c("10569*A", "23189*A", "7272*A_long", "7272
```



Pyrus ussuriensis var. hondoensis

```
ussu1 <- subset(angio_data_long, TreeName == "11728*A")

ussu2 <- subset(angio_data_long, TreeName == "11728*B")

ussu3 <- subset(angio_data_long, TreeName == "11728*C")

ussu4 <- subset(angio_data_long, TreeName == "117456*B")

ussu5 <- subset(angio_data_long, TreeName == "17456*A")

ussu6 <- subset(angio_data_long, TreeName == "17456*B")


plot(ussu1$Year, y = ussu1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(ussu2$Year, y = ussu2$Ringwidth, type = "o", col = "blue")
lines(ussu3$Year, y = ussu3$Ringwidth, type = "o", col = "purple")
lines(ussu4$Year, y = ussu4$Ringwidth, type = "o", col = "lightgreen")
lines(ussu5$Year, y = ussu5$Ringwidth, type = "o", col = "pink")
lines(ussu6$Year, y = ussu6$Ringwidth, type = "o", col = "deeppink")
```

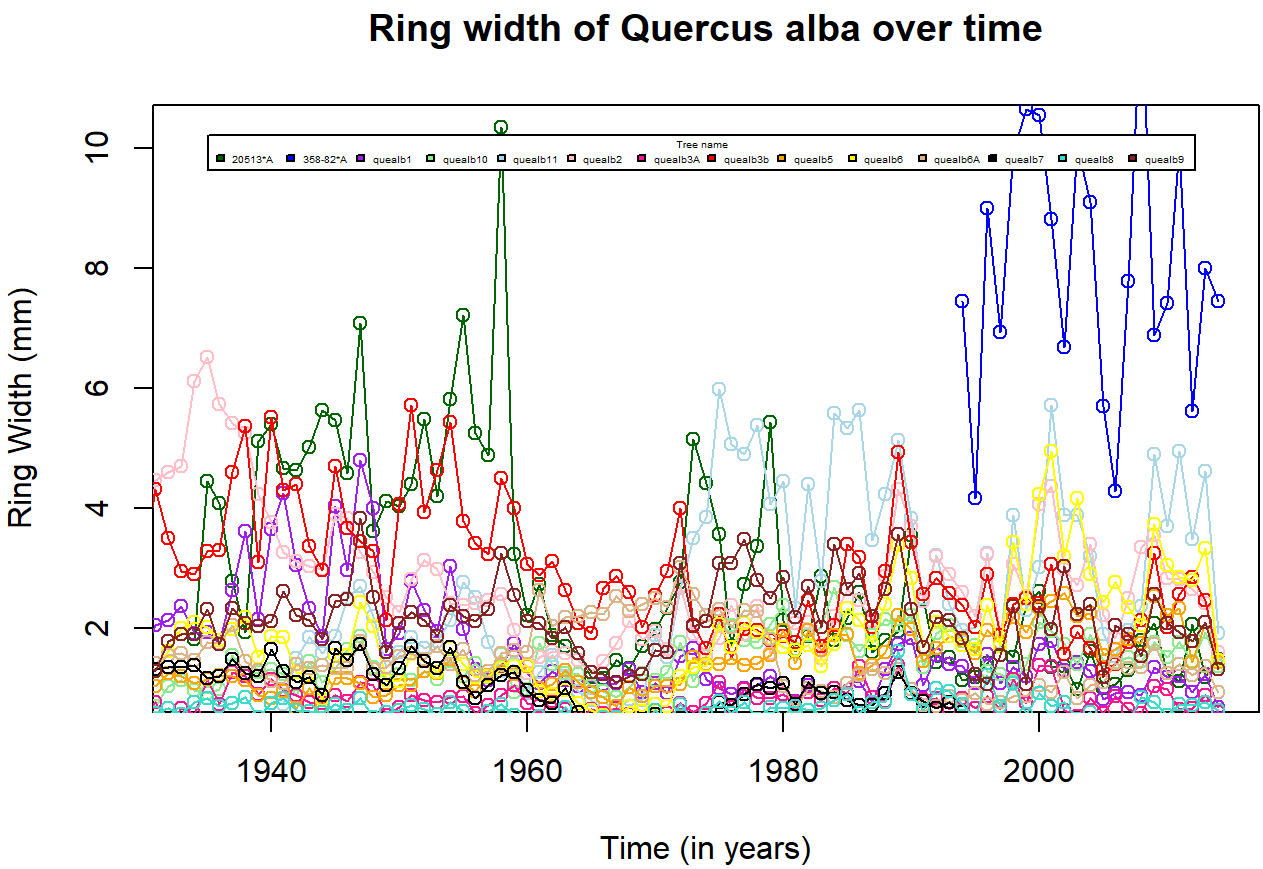
```
legend("topright", inset = .05, title = "Tree name", c("11728*A", "11728*B", "11728*C", "117456*B
```



Quercus alba

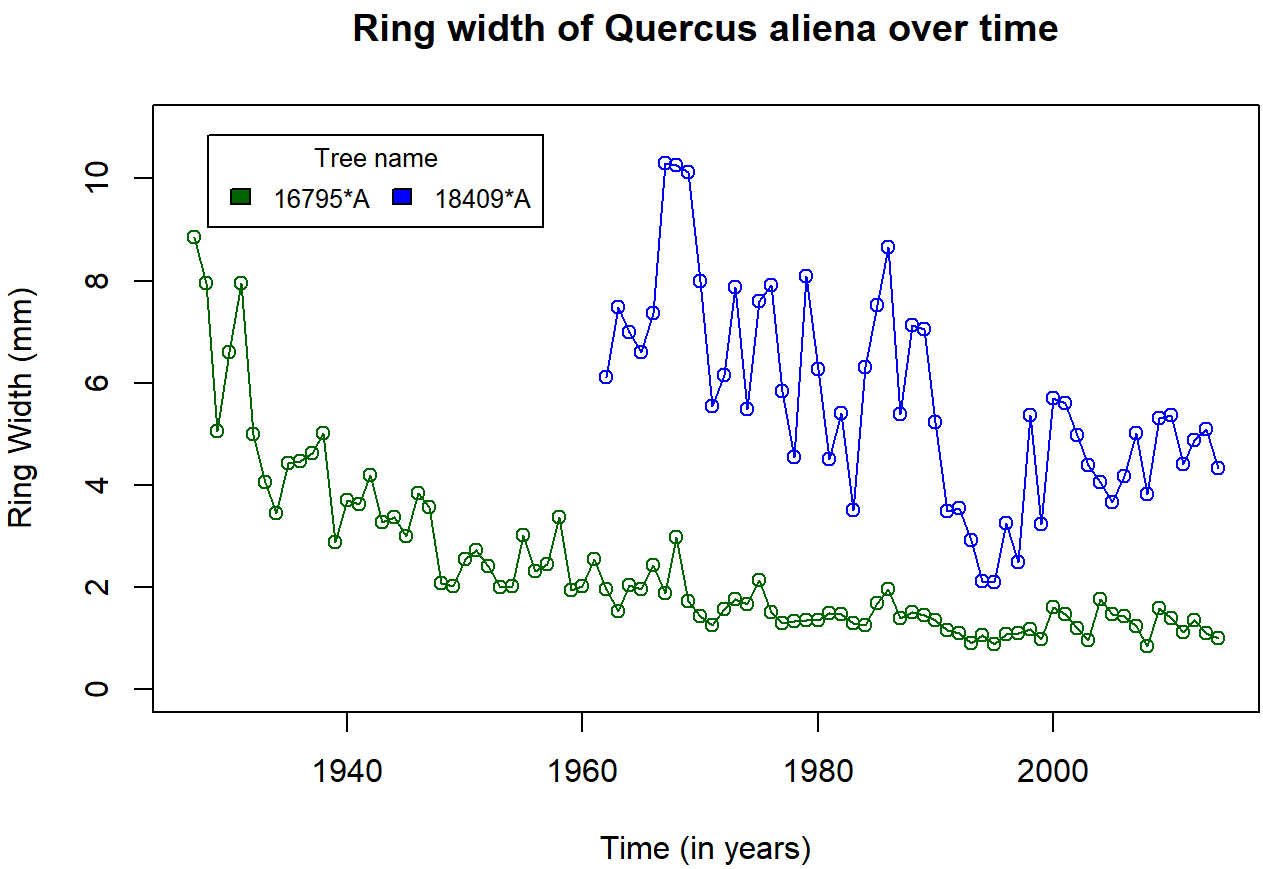
```
alba1 <- subset(angio_data_long, TreeName == "20513*A")

alba2 <- subset(angio_data_long, TreeName == "358-82*A")

alba3 <- subset(angio_data_long, TreeName == "quealb1")

alba4 <- subset(angio_data_long, TreeName == "quealb10")

alba5 <- subset(angio_data_long, TreeName == "quealb11")

alba6 <- subset(angio_data_long, TreeName == "quealb2")

alba7 <- subset(angio_data_long, TreeName == "quealb3A")

alba8 <- subset(angio_data_long, TreeName == "quealb3b")

alba9 <- subset(angio_data_long, TreeName == "quealb5")
```

```
alba10 <- subset(angio_data_long, TreeName == "quealb6")

alba11 <- subset(angio_data_long, TreeName == "quealb6A")

alba12 <- subset(angio_data_long, TreeName == "quealb7")

alba13 <- subset(angio_data_long, TreeName == "quealb8")

alba14 <- subset(angio_data_long, TreeName == "quealb9")


plot(alba1$Year, y = alba1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(alba2$Year, y = alba2$Ringwidth, type = "o", col = "blue")
lines(alba3$Year, y = alba3$Ringwidth, type = "o", col = "purple")
lines(alba4$Year, y = alba4$Ringwidth, type = "o", col = "lightgreen")
lines(alba5$Year, y = alba5$Ringwidth, type = "o", col = "lightblue")
lines(alba6$Year, y = alba6$Ringwidth, type = "o", col = "pink")
lines(alba7$Year, y = alba7$Ringwidth, type = "o", col = "deeppink")
lines(alba8$Year, y = alba8$Ringwidth, type = "o", col = "red")
lines(alba9$Year, y = alba9$Ringwidth, type = "o", col = "orange")
lines(alba10$Year, y = alba10$Ringwidth, type = "o", col = "yellow")
lines(alba11$Year, y = alba11$Ringwidth, type = "o", col = "burlywood")
lines(alba12$Year, y = alba12$Ringwidth, type = "o", col = "black")
lines(alba13$Year, y = alba13$Ringwidth, type = "o", col = "turquoise")
lines(alba14$Year, y = alba14$Ringwidth, type = "o", col = "brown4")

legend("topleft", inset = .05, title = "Tree name", c("20513*A", "358-82*A", "quealb1", "quealb10
```



Quercus aliena

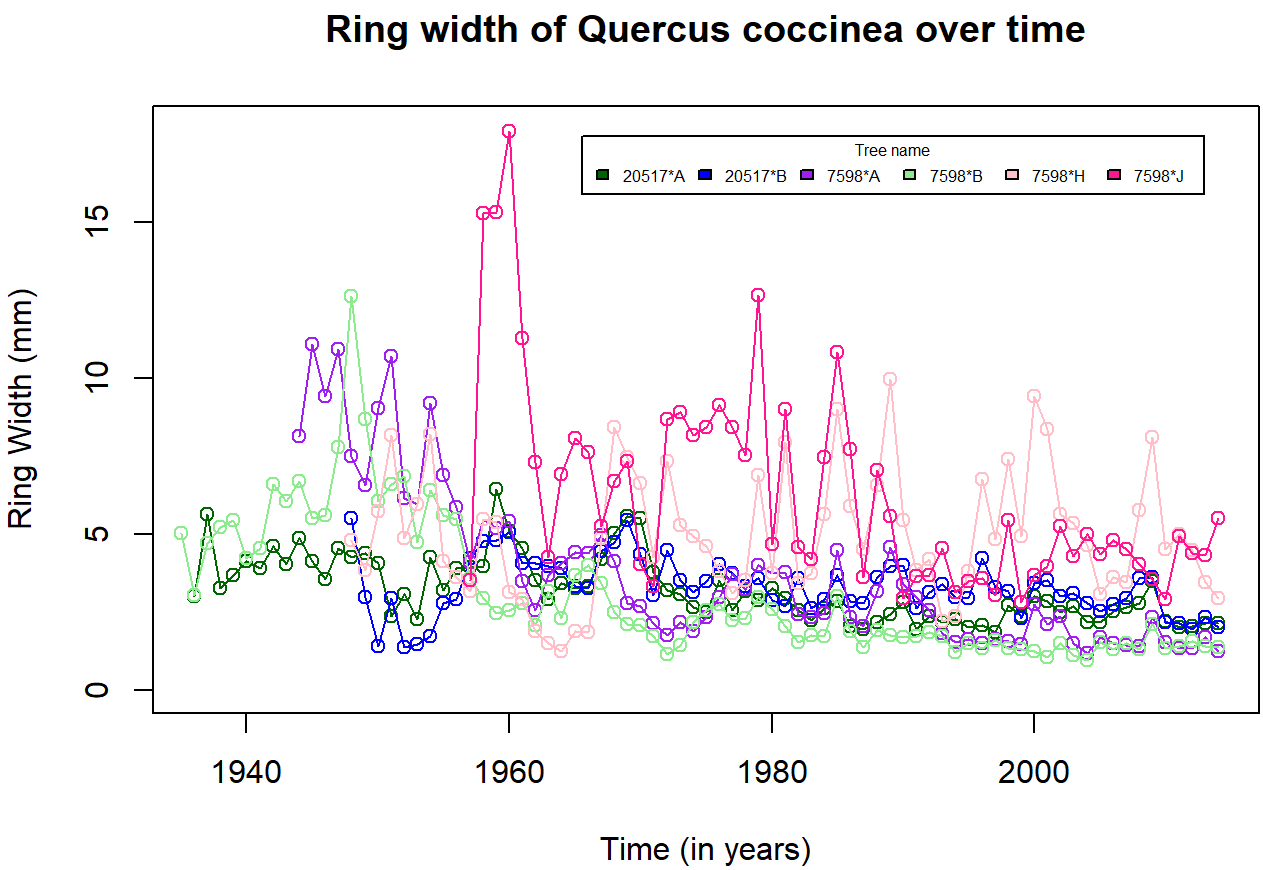
```
alien1 <- subset(angio_data_long, TreeName == "16795*A")

alien2 <- subset(angio_data_long, TreeName == "18409*A")


plot(alien1$Year, y = alien1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(alien2$Year, y = alien2$Ringwidth, type = "o", col = "blue")

legend("topleft", inset = .05, title = "Tree name", c("16795*A", "18409*A"), fill = c("darkgreen",
```



Quercus coccinea

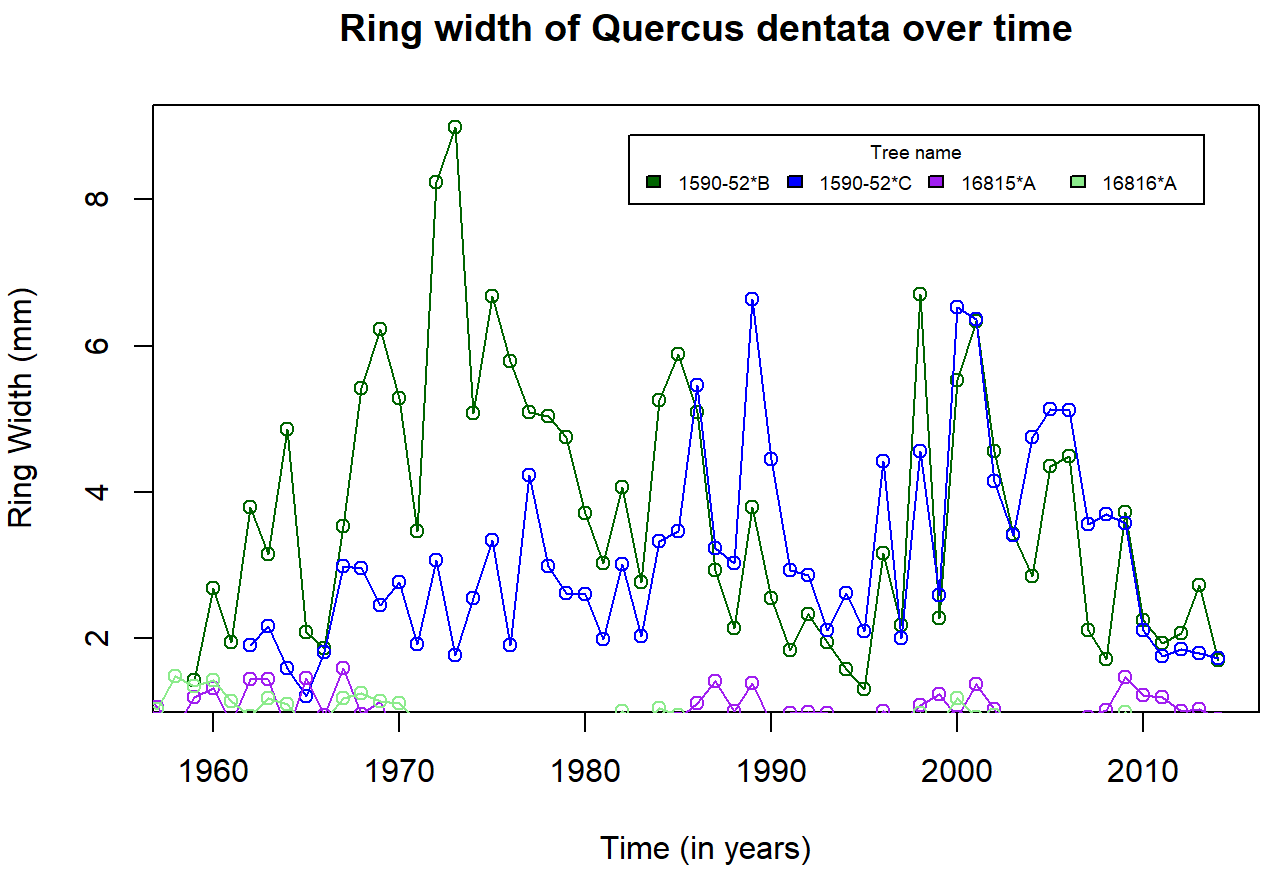
```
cocci1 <- subset(angio_data_long, TreeName == "20517*A")

cocci2 <- subset(angio_data_long, TreeName == "20517*B")

cocci3 <- subset(angio_data_long, TreeName == "7598*A")

cocci4 <- subset(angio_data_long, TreeName == "7598*B")

cocci5 <- subset(angio_data_long, TreeName == "7598*H")

cocci6 <- subset(angio_data_long, TreeName == "7598*J")


plot(cocci1$Year, y = cocci1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(cocci2$Year, y = cocci2$Ringwidth, type = "o", col = "blue")
lines(cocci3$Year, y = cocci3$Ringwidth, type = "o", col = "purple")
lines(cocci4$Year, y = cocci4$Ringwidth, type = "o", col = "lightgreen")
lines(cocci5$Year, y = cocci5$Ringwidth, type = "o", col = "pink")
lines(cocci6$Year, y = cocci6$Ringwidth, type = "o", col = "deeppink")
```

```
legend("topright", inset = .05, title = "Tree name", c("20517*A", "20517*B", "7598*A", "7598*B",
```



Quercus dentata

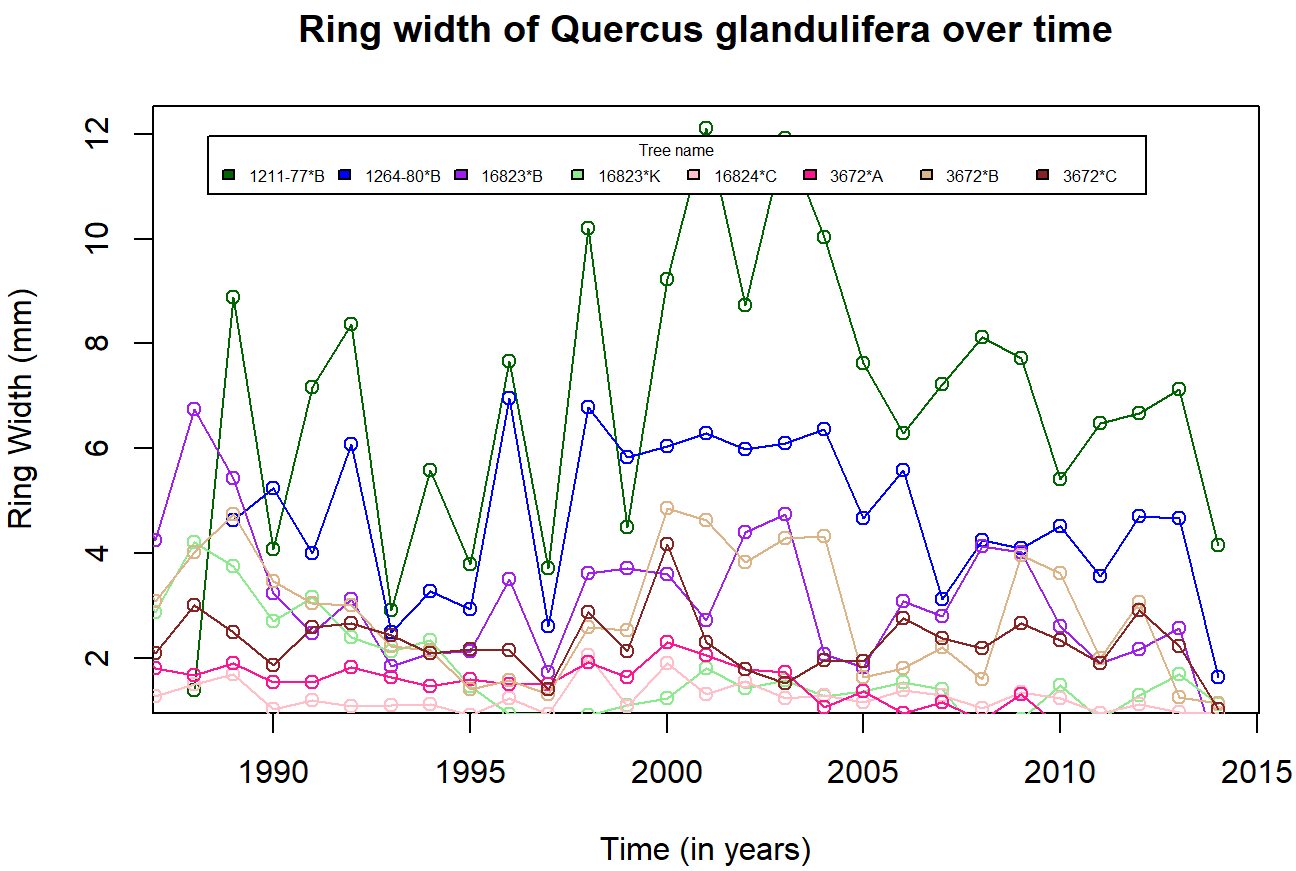
```
dent1 <- subset(angio_data_long, TreeName == "1590-52*B")

dent2 <- subset(angio_data_long, TreeName == "1590-52*C")

dent3 <- subset(angio_data_long, TreeName == "16815*A")

dent4 <- subset(angio_data_long, TreeName == "16816*A")


plot(dent1$Year, y = dent1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab
lines(dent2$Year, y = dent2$Ringwidth, type = "o", col = "blue")
lines(dent3$Year, y = dent3$Ringwidth, type = "o", col = "purple")
lines(dent4$Year, y = dent4$Ringwidth, type = "o", col = "lightgreen")

legend("topright", inset = .05, title = "Tree name", c("1590-52*B", "1590-52*C", "16815*A", "1681
```



Quercus glandulifera

```
glandu1 <- subset(angio_data_long, TreeName == "1211-77*B")

glandu2 <- subset(angio_data_long, TreeName == "1264-80*B")

glandu3 <- subset(angio_data_long, TreeName == "16823*B")

glandu4 <- subset(angio_data_long, TreeName == "16823*K")

glandu5 <- subset(angio_data_long, TreeName == "16824*C")

glandu6 <- subset(angio_data_long, TreeName == "3672*A")

glandu7 <- subset(angio_data_long, TreeName == "3672*B")

glandu8 <- subset(angio_data_long, TreeName == "3672*C")


plot(glandu1$Year, y = glandu1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", y
lines(glandu2$Year, y = glandu2$Ringwidth, type = "o", col = "blue")
lines(glandu3$Year, y = glandu3$Ringwidth, type = "o", col = "purple")
lines(glandu4$Year, y = glandu4$Ringwidth, type = "o", col = "lightgreen")
```

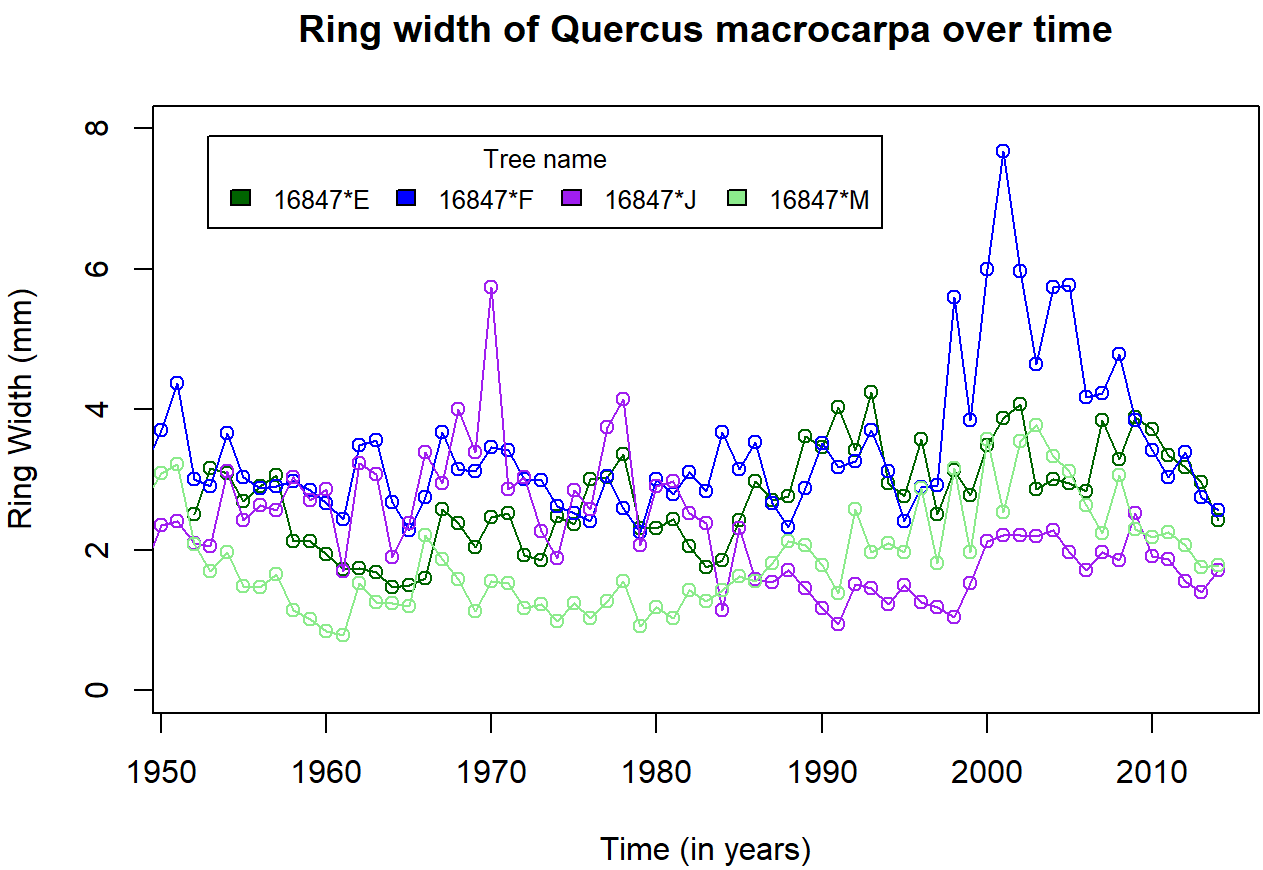
```
lines(glandu5$Year, y = glandu5$Ringwidth, type = "o", col = "pink")
lines(glandu6$Year, y = glandu6$Ringwidth, type = "o", col = "deeppink")
lines(glandu7$Year, y = glandu7$Ringwidth, type = "o", col = "burlywood")
lines(glandu8$Year, y = glandu8$Ringwidth, type = "o", col = "brown4")

legend("topleft", inset = .05, title = "Tree name", c("1211-77*B", "1264-80*B", "16823*B", "16823
```



Quercus macrocarpa

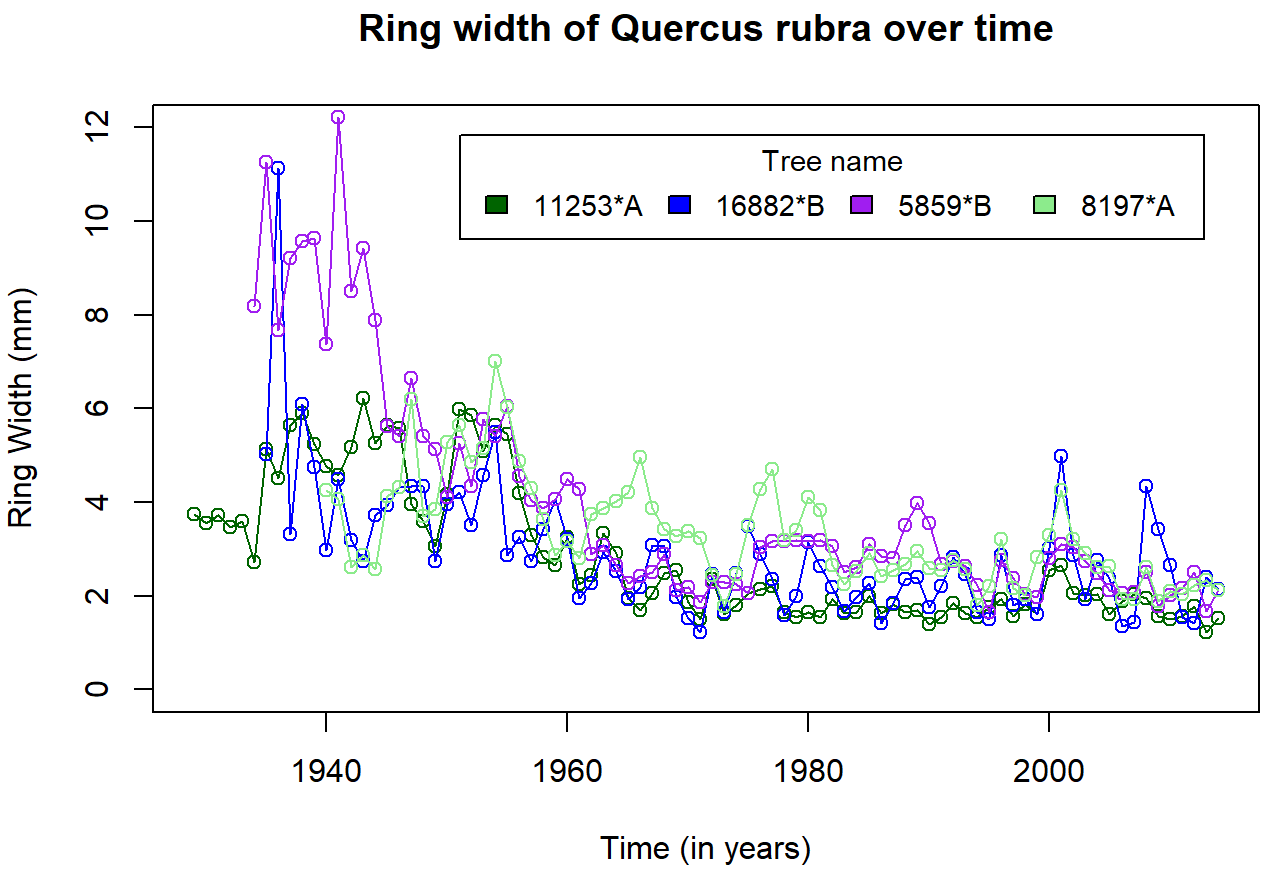
```
macro1 <- subset(angio_data_long, TreeName == "16847*E")

macro2 <- subset(angio_data_long, TreeName == "16847*F")

macro3 <- subset(angio_data_long, TreeName == "16847*J")

macro4 <- subset(angio_data_long, TreeName == "16847*M")


plot(macro1$Year, y = macro1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(macro2$Year, y = macro2$Ringwidth, type = "o", col = "blue")
lines(macro3$Year, y = macro3$Ringwidth, type = "o", col = "purple")
lines(macro4$Year, y = macro4$Ringwidth, type = "o", col = "lightgreen")
```

```
legend("topleft", inset = .05, title = "Tree name", c("16847*E", "16847*F", "16847*J", "16847*M")
```



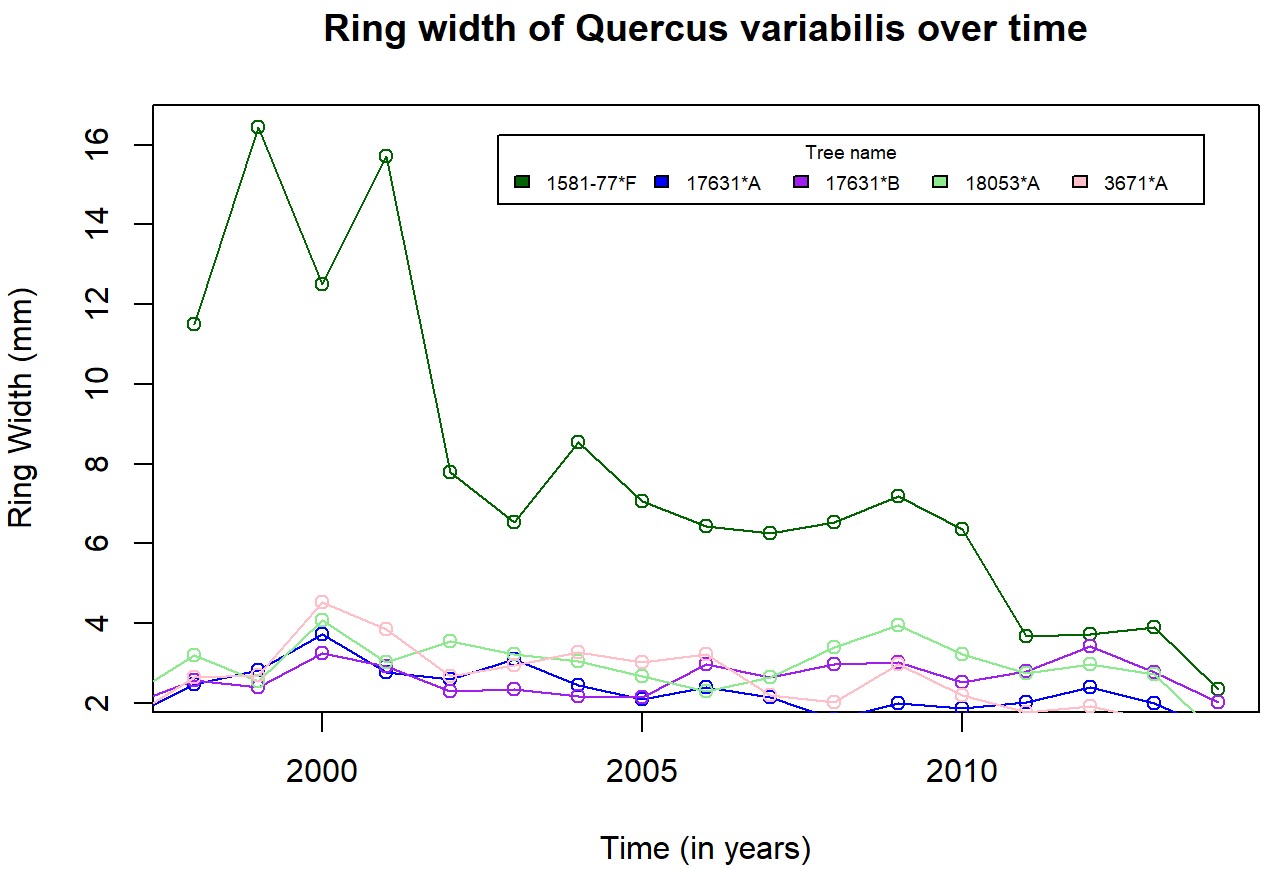
Quercus rubra

```
rubra1 <- subset(angio_data_long, TreeName == "11253*A")

rubra2 <- subset(angio_data_long, TreeName == "16882*B")

rubra3 <- subset(angio_data_long, TreeName == "5859*B")

rubra4 <- subset(angio_data_long, TreeName == "8197*A")


plot(rubra1$Year, y = rubra1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(rubra2$Year, y = rubra2$Ringwidth, type = "o", col = "blue")
lines(rubra3$Year, y = rubra3$Ringwidth, type = "o", col = "purple")
lines(rubra4$Year, y = rubra4$Ringwidth, type = "o", col = "lightgreen")

legend("topright", inset = .05, title = "Tree name", c("11253*A", "16882*B", "5859*B", "8197*A"),
```



Quercus variabilis

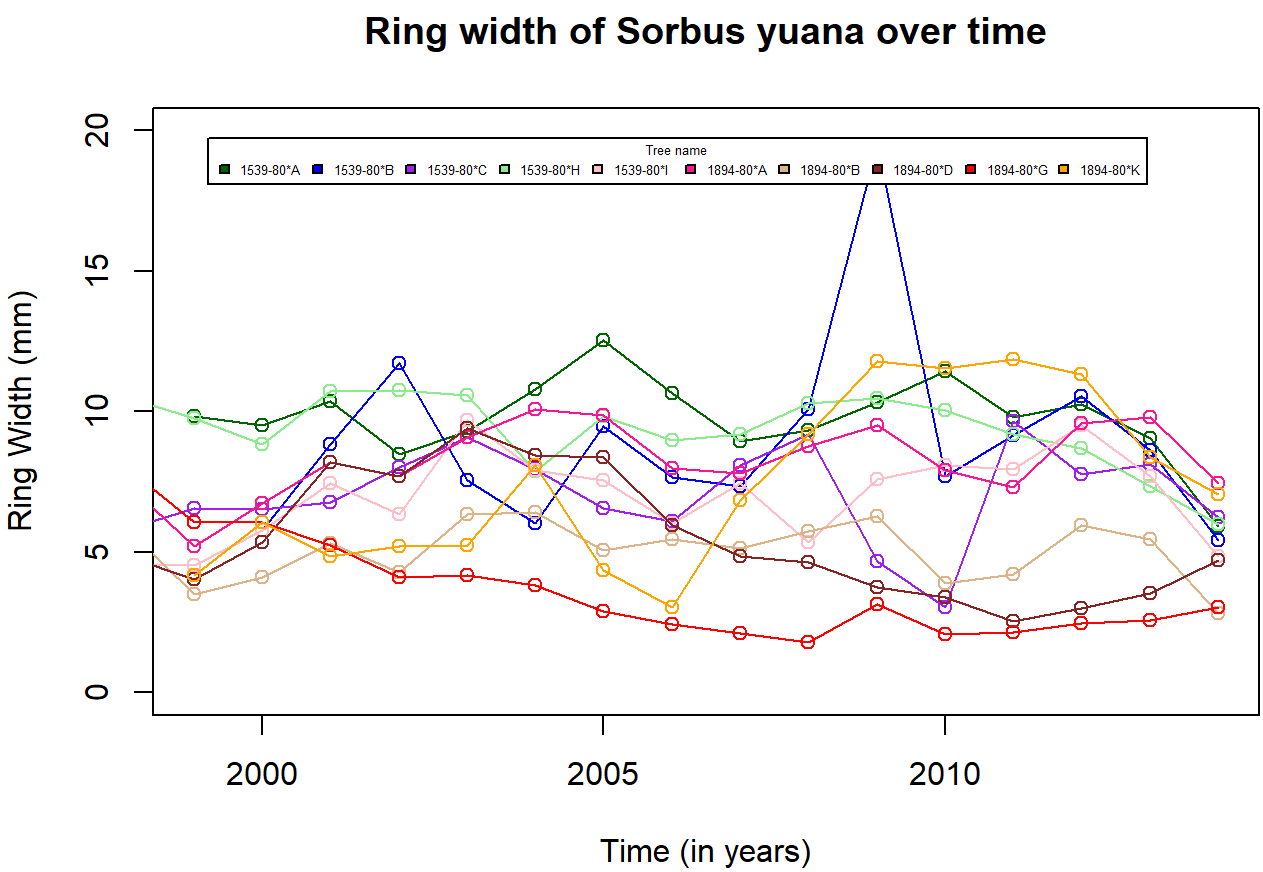
```
varia1 <- subset(angio_data_long, TreeName == "1581-77*F")

varia2 <- subset(angio_data_long, TreeName == "17631*A")

varia3 <- subset(angio_data_long, TreeName == "17631*B")

varia4 <- subset(angio_data_long, TreeName == "18053*A")

varia5 <- subset(angio_data_long, TreeName == "3671*A")


plot(varia1$Year, y = varia1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(varia2$Year, y = varia2$Ringwidth, type = "o", col = "blue")
lines(varia3$Year, y = varia3$Ringwidth, type = "o", col = "purple")
lines(varia4$Year, y = varia4$Ringwidth, type = "o", col = "lightgreen")
lines(varia5$Year, y = varia5$Ringwidth, type = "o", col = "pink")

legend("topright", inset = .05, title = "Tree name", c("1581-77*F", "17631*A", "17631*B", "18053*A
```



Sorbus yuana

```
yuana1 <- subset(angio_data_long, TreeName == "1539-80*A")

yuana2 <- subset(angio_data_long, TreeName == "1539-80*B")

yuana3 <- subset(angio_data_long, TreeName == "1539-80*C")

yuana4 <- subset(angio_data_long, TreeName == "1539-80*H")

yuana5 <- subset(angio_data_long, TreeName == "1539-80*I")

yuana6 <- subset(angio_data_long, TreeName == "1894-80*A")

yuana7 <- subset(angio_data_long, TreeName == "1894-80*B")

yuana8 <- subset(angio_data_long, TreeName == "1894-80*D")

yuana9 <- subset(angio_data_long, TreeName == "1894-80*G")

yuana10 <- subset(angio_data_long, TreeName == "1894-80*K")
```

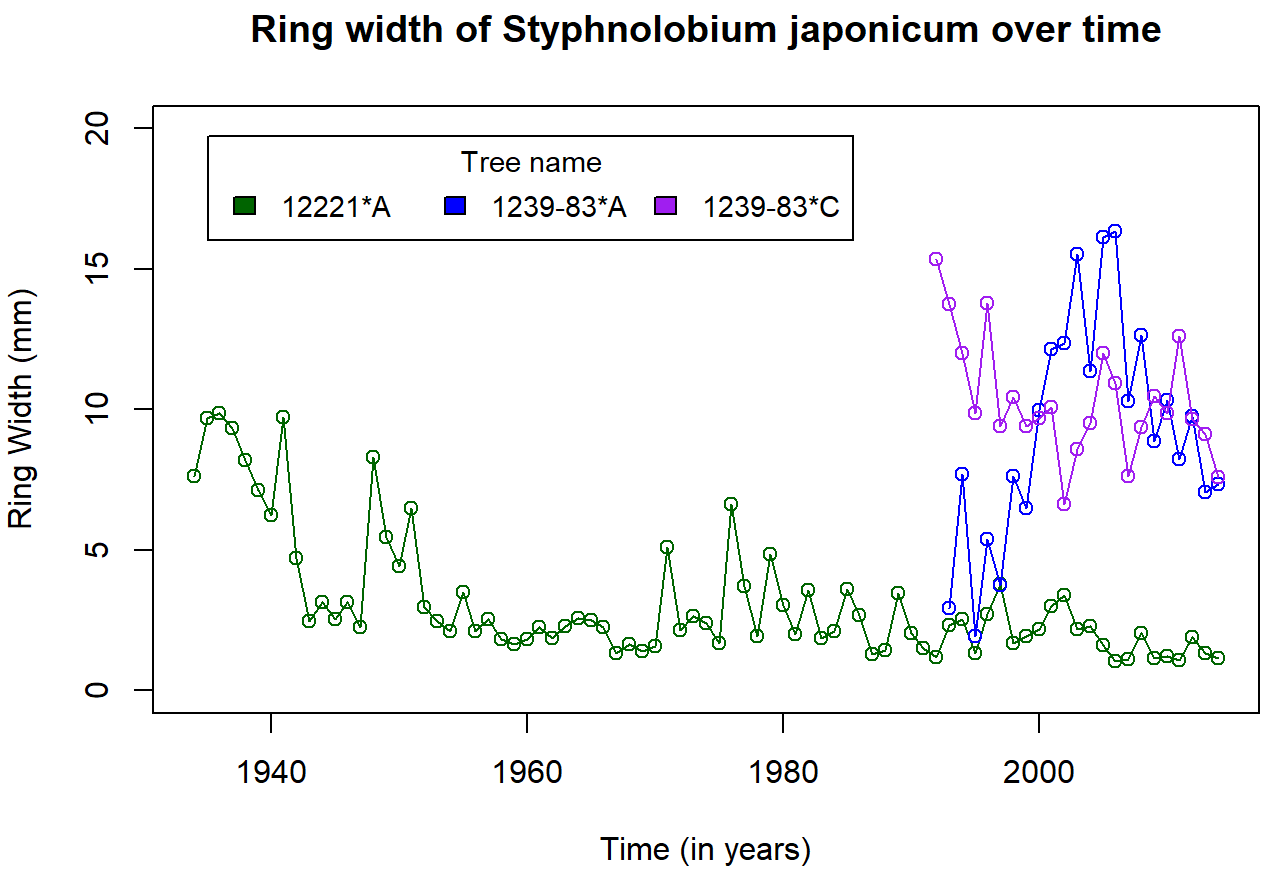
```
plot(yuana1$Year, y = yuana1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(yuana2$Year, y = yuana2$Ringwidth, type = "o", col = "blue")
lines(yuana3$Year, y = yuana3$Ringwidth, type = "o", col = "purple")
lines(yuana4$Year, y = yuana4$Ringwidth, type = "o", col = "lightgreen")
lines(yuana5$Year, y = yuana5$Ringwidth, type = "o", col = "pink")
lines(yuana6$Year, y = yuana6$Ringwidth, type = "o", col = "deeppink")
lines(yuana7$Year, y = yuana7$Ringwidth, type = "o", col = "burlywood")
lines(yuana8$Year, y = yuana8$Ringwidth, type = "o", col = "brown4")
lines(yuana9$Year, y = yuana9$Ringwidth, type = "o", col = "red")
lines(yuana10$Year, y = yuana10$Ringwidth, type = "o", col = "orange")


legend("topleft", inset = .05, title = "Tree name", c("1539-80*A", "1539-80*B", "1539-80*C", "153
```



Styphnolobium japonicum

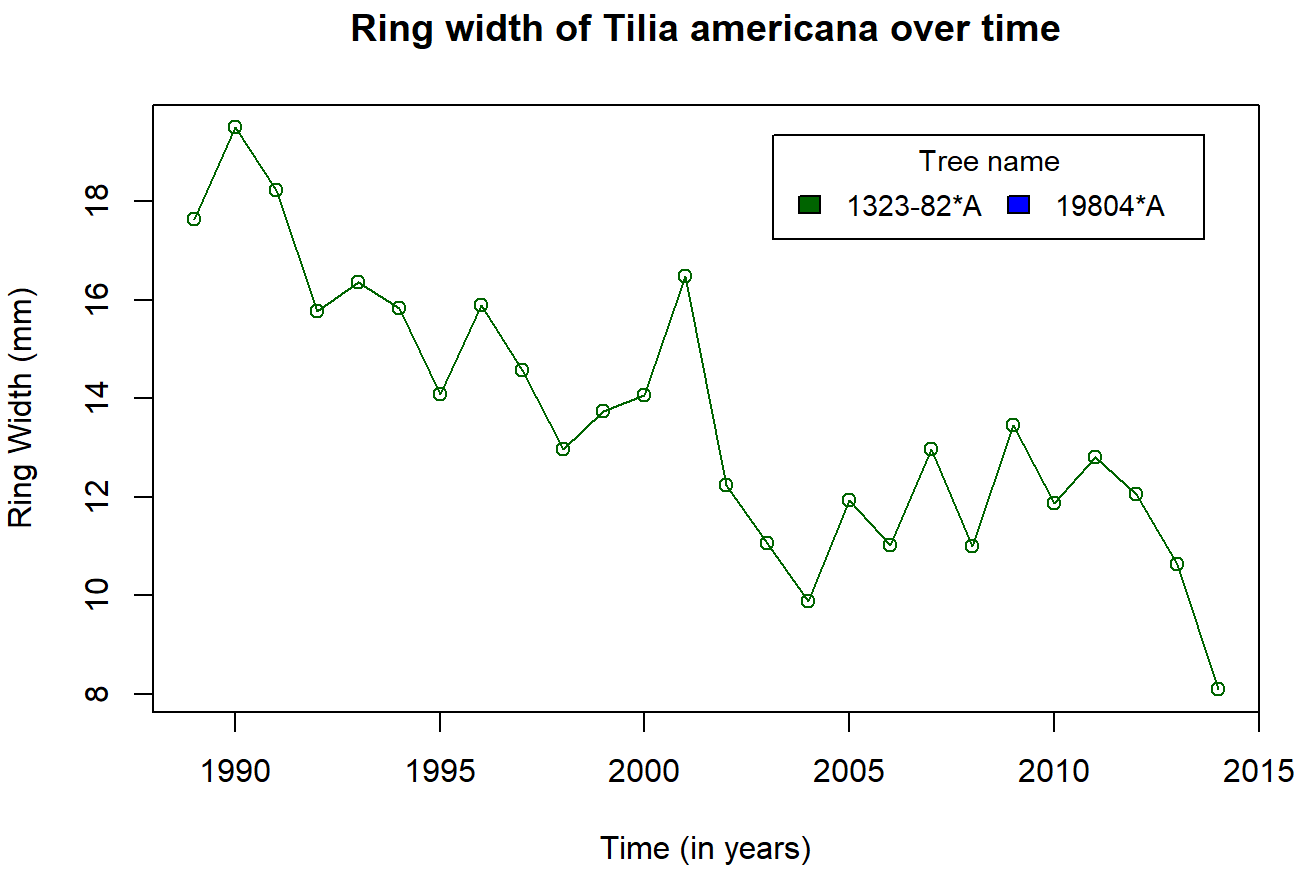
```
styph1 <- subset(angio_data_long, TreeName == "12221*A")

styph2 <- subset(angio_data_long, TreeName == "1239-83*A")

styph3 <- subset(angio_data_long, TreeName == "1239-83*C")
```

```
plot(styph1$Year, y = styph1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(styph2$Year, y = styph2$Ringwidth, type = "o", col = "blue")
lines(styph3$Year, y = styph3$Ringwidth, type = "o", col = "purple")


legend("topleft", inset = .05, title = "Tree name", c("12221*A", "1239-83*A", "1239-83*C"), fill =
```



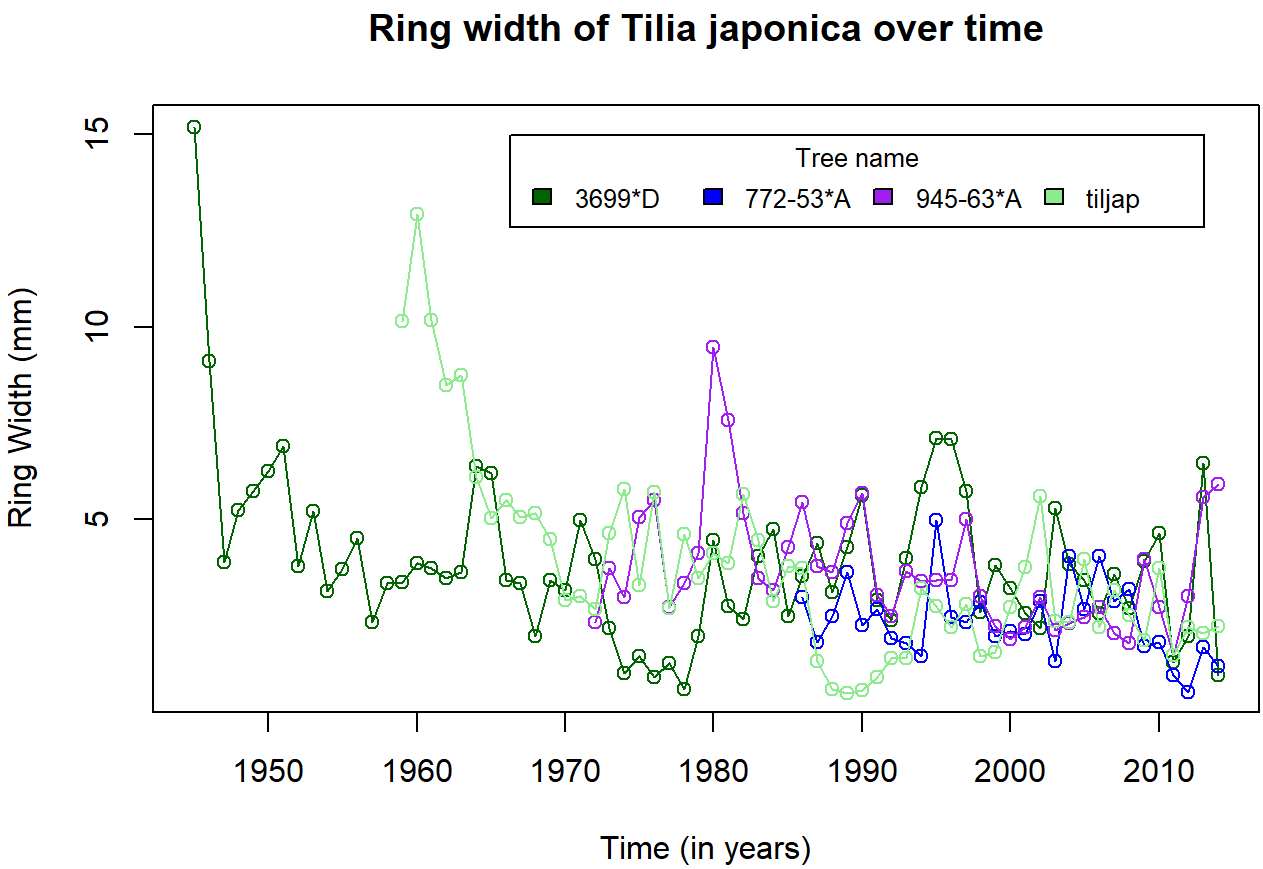
Tilia americana

```
ameri1 <- subset(angio_data_long, TreeName == "1323-82*A")

ameri2 <- subset(angio_data_long, TreeName == "19804*A")


plot(ameri1$Year, y = ameri1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(ameri2$Year, y = ameri2$Ringwidth, type = "o", col = "blue")

legend("topright", inset = .05, title = "Tree name", c("1323-82*A", "19804*A"), fill = c("darkgre
```



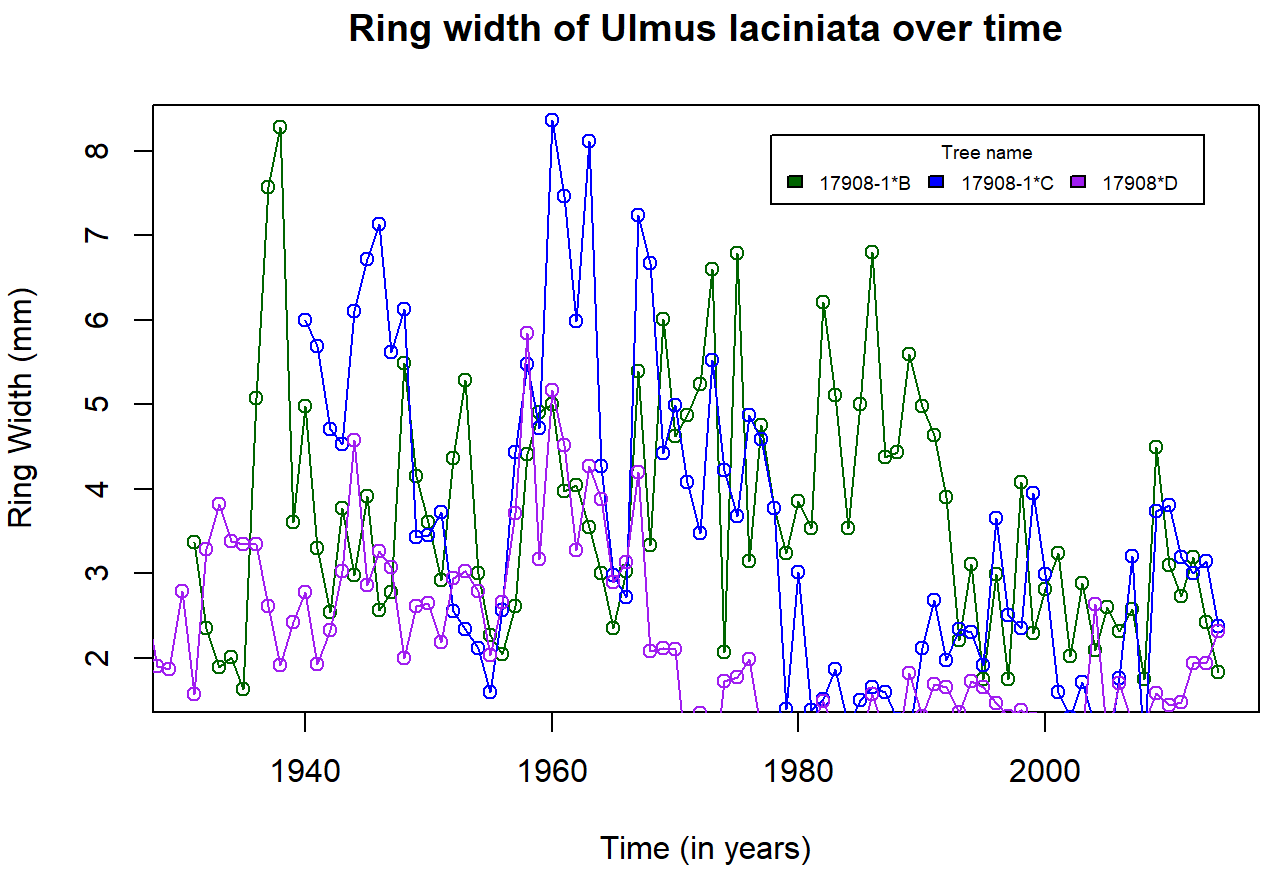
Tilia japonica

```
japon1 <- subset(angio_data_long, TreeName == "3699*D")

japon2 <- subset(angio_data_long, TreeName == "772-53*A")

japon3 <- subset(angio_data_long, TreeName == "945-63*A")

japon4 <- subset(angio_data_long, TreeName == "tiljap")


plot(japon1$Year, y = japon1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylal
lines(japon2$Year, y = japon2$Ringwidth, type = "o", col = "blue")
lines(japon3$Year, y = japon3$Ringwidth, type = "o", col = "purple")
lines(japon4$Year, y = japon4$Ringwidth, type = "o", col = "lightgreen")

legend("topright", inset = .05, title = "Tree name", c("3699*D", "772-53*A", "945-63*A", "tiljap"
```

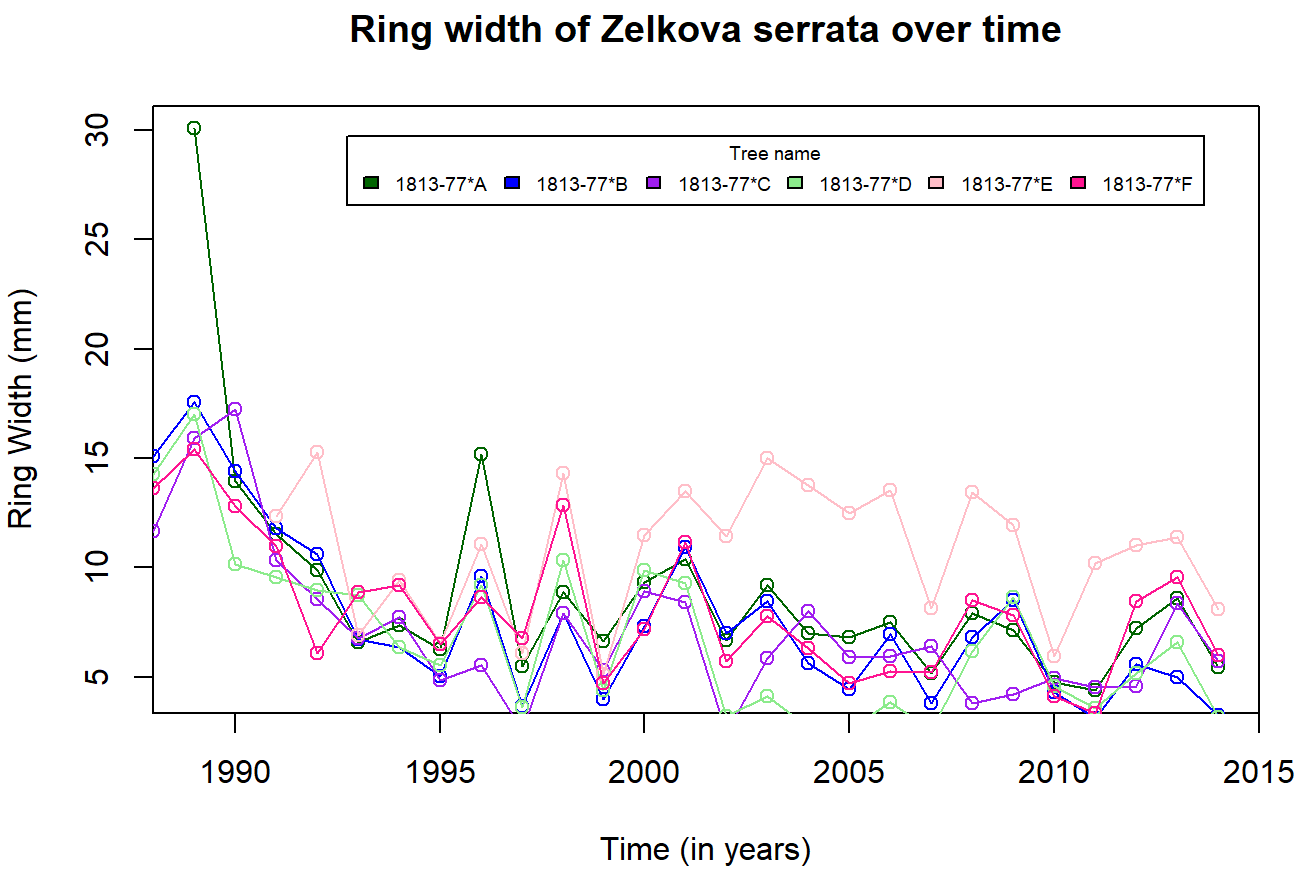


Ulmus laciniata

```
laci1 <- subset(angio_data_long, TreeName == "17908-1*B")

laci2 <- subset(angio_data_long, TreeName == "17908-1*C")

laci3 <- subset(angio_data_long, TreeName == "17908*D")


plot(laci1$Year, y = laci1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(laci2$Year, y = laci2$Ringwidth, type = "o", col = "blue")
lines(laci3$Year, y = laci3$Ringwidth, type = "o", col = "purple")

legend("topright", inset = .05, title = "Tree name", c("17908-1*B", "17908-1*C", "17908*D"), fill
```



Zelkova serrata

```
zelk1 <- subset(angio_data_long, TreeName == "1813-77*A")

zelk2 <- subset(angio_data_long, TreeName == "1813-77*B")

zelk3 <- subset(angio_data_long, TreeName == "1813-77*C")

zelk4 <- subset(angio_data_long, TreeName == "1813-77*D")

zelk5 <- subset(angio_data_long, TreeName == "1813-77*E")

zelk6 <- subset(angio_data_long, TreeName == "1813-77*F")


plot(zelk1$Year, y = zelk1$Ringwidth, type = "o", col="darkgreen", xlab= "Time (in years)", ylab =
lines(zelk2$Year, y = zelk2$Ringwidth, type = "o", col = "blue")
lines(zelk3$Year, y = zelk3$Ringwidth, type = "o", col = "purple")
lines(zelk4$Year, y = zelk4$Ringwidth, type = "o", col = "lightgreen")
lines(zelk5$Year, y = zelk5$Ringwidth, type = "o", col = "pink")
lines(zelk6$Year, y = zelk6$Ringwidth, type = "o", col = "deeppink")
```

```
legend("topright", inset = .05, title = "Tree name", c("1813-77*A", "1813-77*B", "1813-77*C", "181
```